



**FPT SCHOOL OF BUSINESS
& TECHNOLOGY**

IMPLEMENTATION OF DISCRETE-TIME SYSTEMS

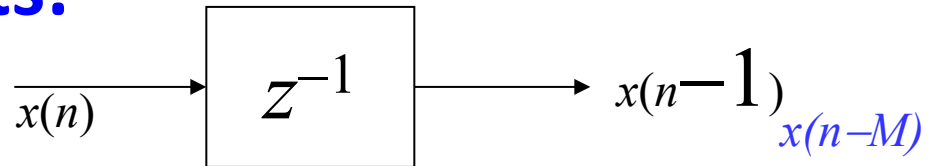
- Understand the block diagram of systems.
- Understand basic structures of IIR and FIR
- Understand the effects of quantization

1. Block Diagram Representation of Linear Constant-Coefficient Difference Equations
2. Basic Structures for IIR Systems
3. Basic Structures for FIR Systems
4. The Effects of Coefficient Quantization

$$y(n) = ay(n-1) + b_0x(n) + b_1x(n-1)$$

Three basic elements:

Unit Delay

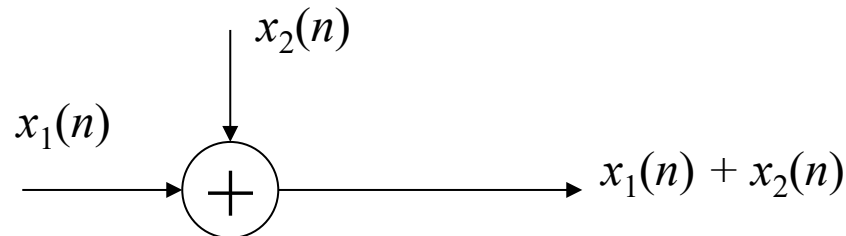


(Memory, storage) M sample Delay z^{-M} 

Multiplier



Adder

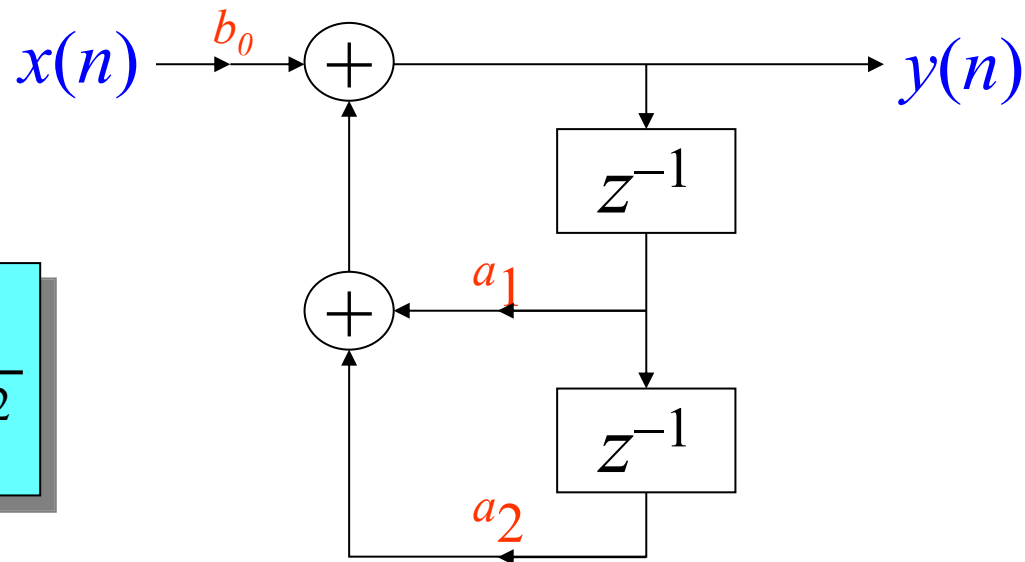


- Example: Draw Block Diagram Representation of a Second-order Difference Equation

$$y(n] = a_1 y(n-1) + a_2 y(n-2) + b_0 x(n)$$

Solution:

$$H(z) = \frac{b_0}{1 - a_1 z^{-1} - a_2 z^{-2}}$$



- Higher-Order Difference Equations

$$\sum_{k=0}^N a_k y(n-k) = \sum_{k=0}^M b_k x(n-k)$$

$$y(n) - \sum_{k=1}^N a_k y(n-k) = \sum_{k=0}^M b_k x(n-k)$$

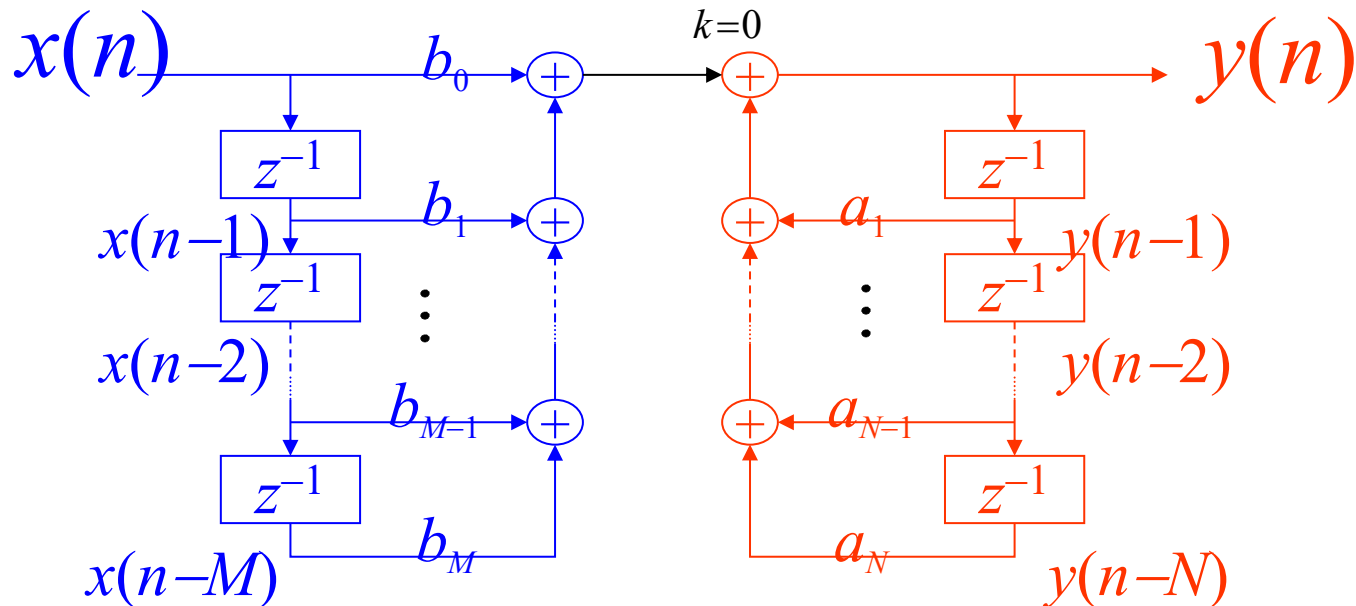
$$y(n) = \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

$$H(z) = \frac{\sum_{k=1}^M b_k z^{-k}}{1 - \sum_{k=1}^N a_k z^{-k}}$$

- Block Diagram Representation (Direct Form I)

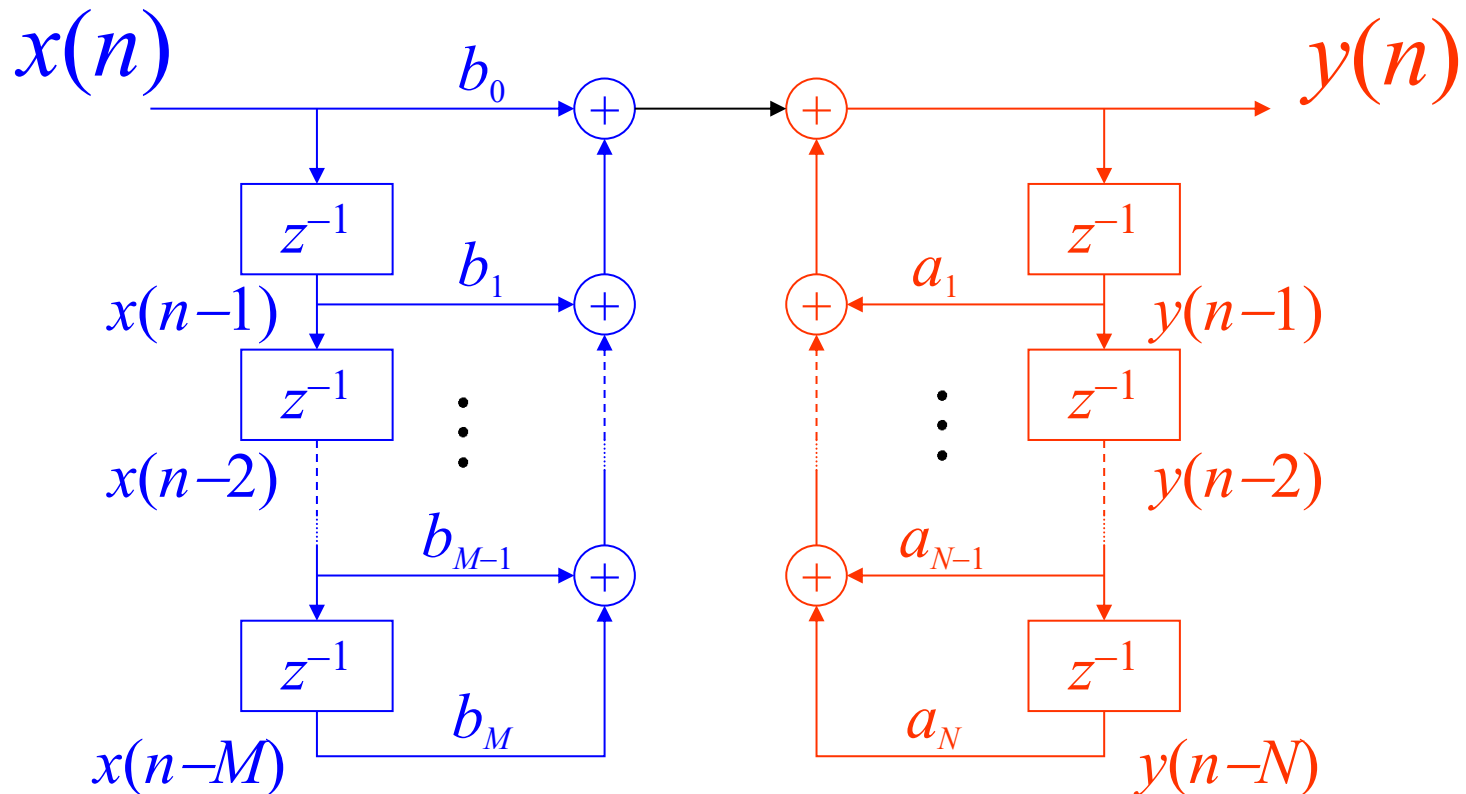
$$y(n) = \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k) \quad y(n) = \sum_{k=1}^N a_k y(n-k) + v(n)$$

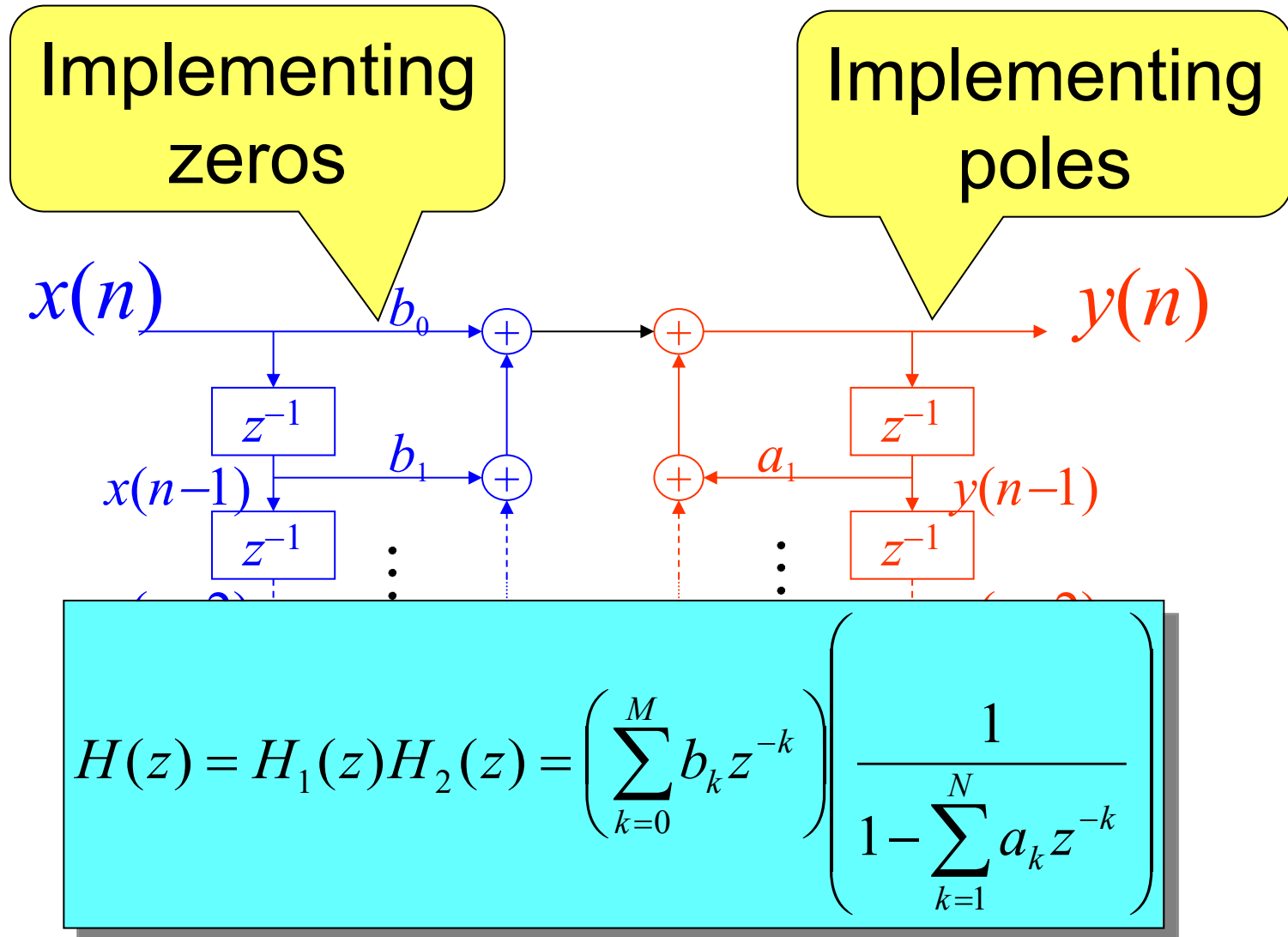
$$v(n) = \sum_{k=0}^M b_k x(n-k)$$



$$v(n) = \sum_{k=0}^M b_k x(n-k)$$

$$y(n) = \sum_{k=1}^N a_k y(n-k) + v(n)$$



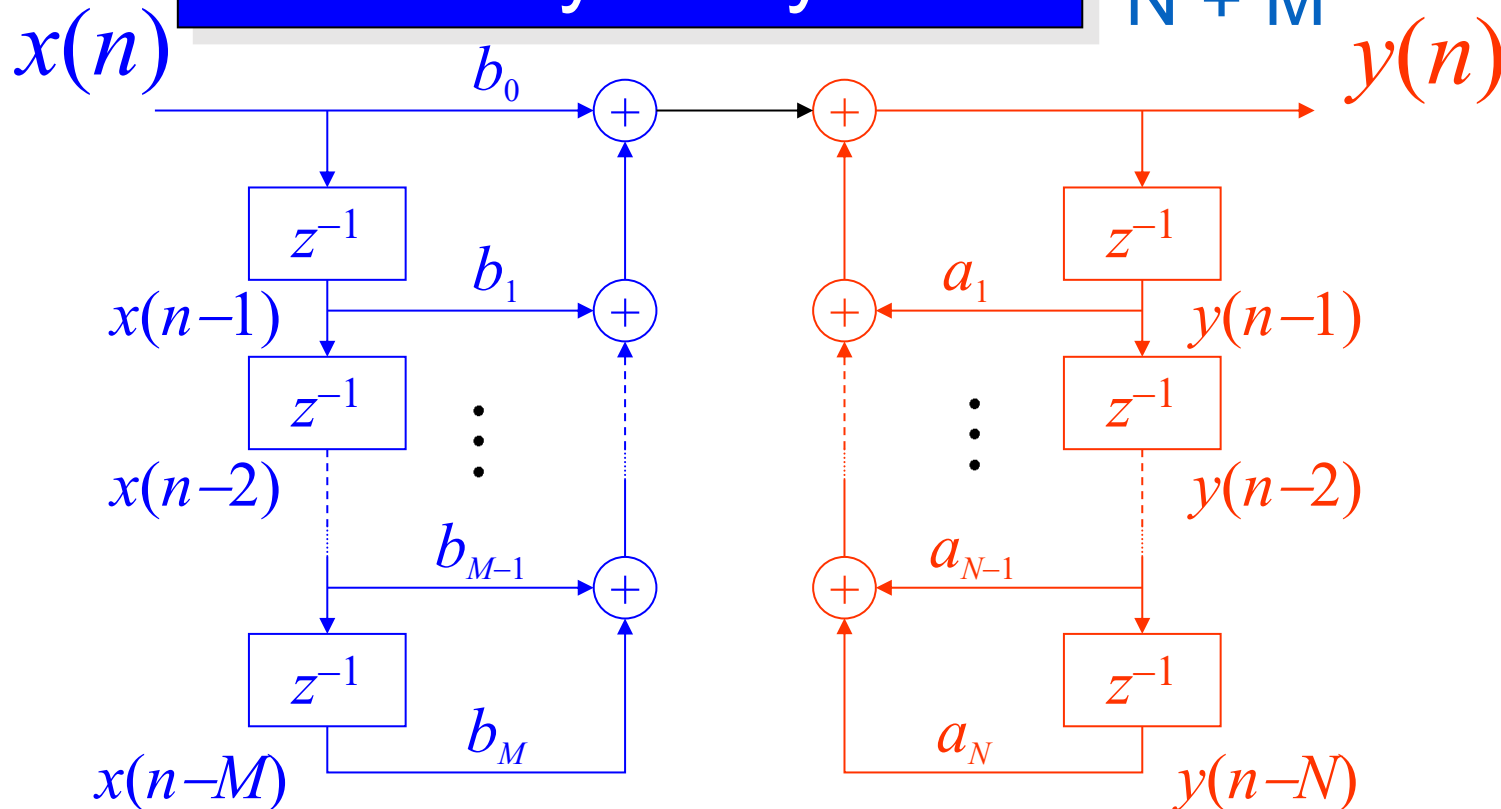


How many Adders?
How many multipliers?
How many delays?

$$N + M$$

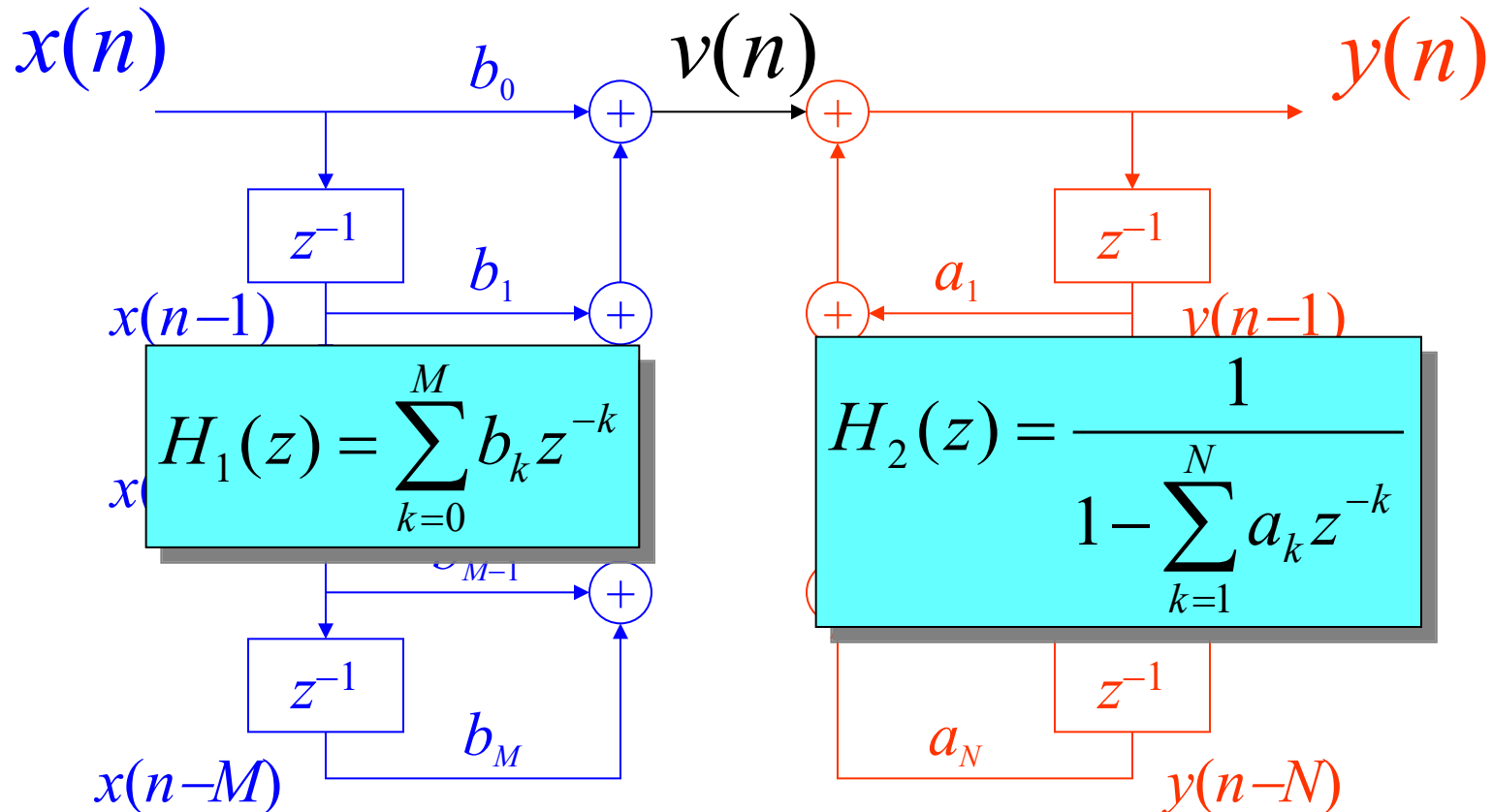
$$N + M + 1$$

$$N + M$$

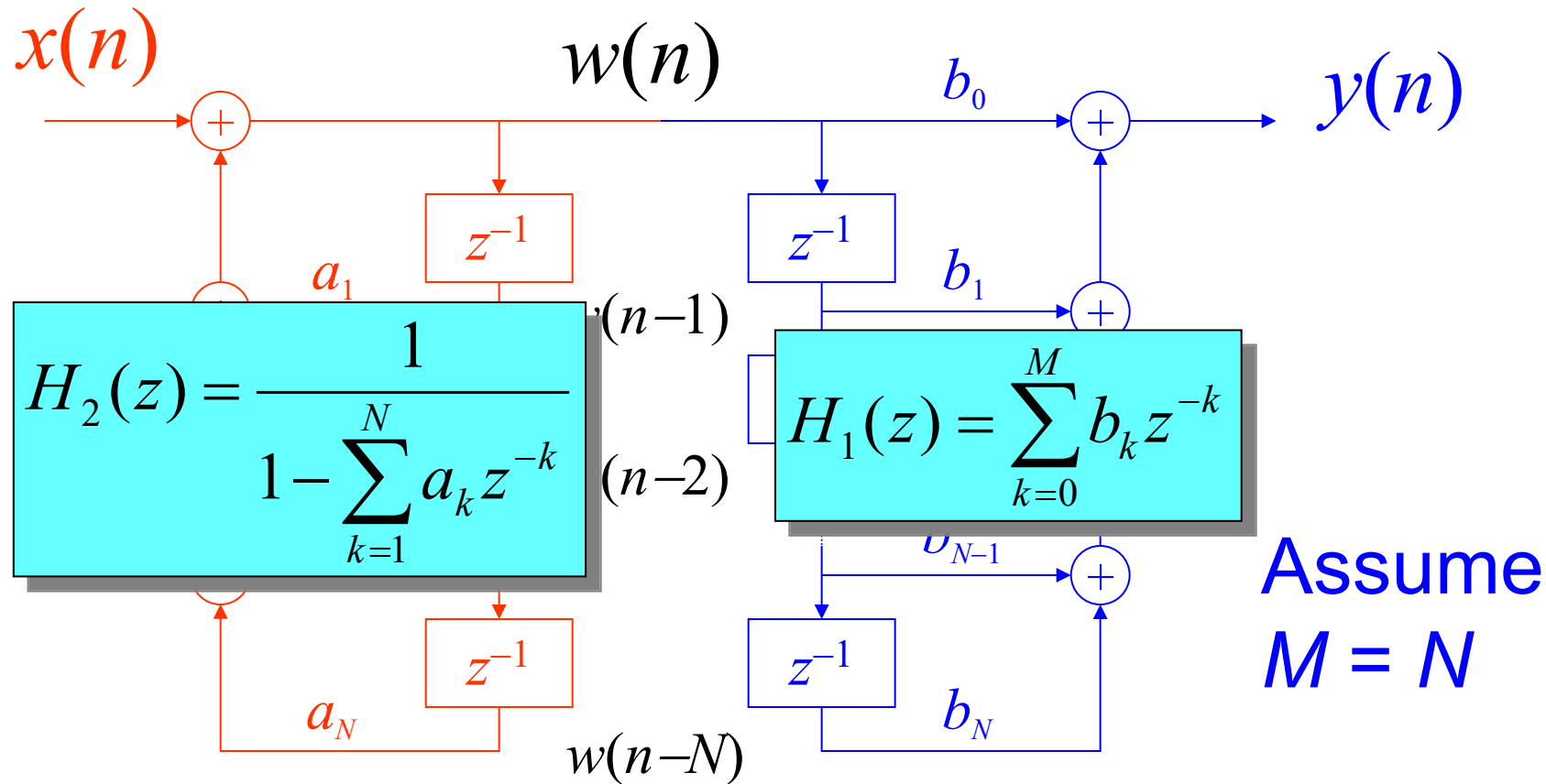


$$v(n) = \sum_{k=0}^M b_k x(n-k)$$

$$y(n) = \sum_{k=1}^N a_k y(n-k) + v(n)$$

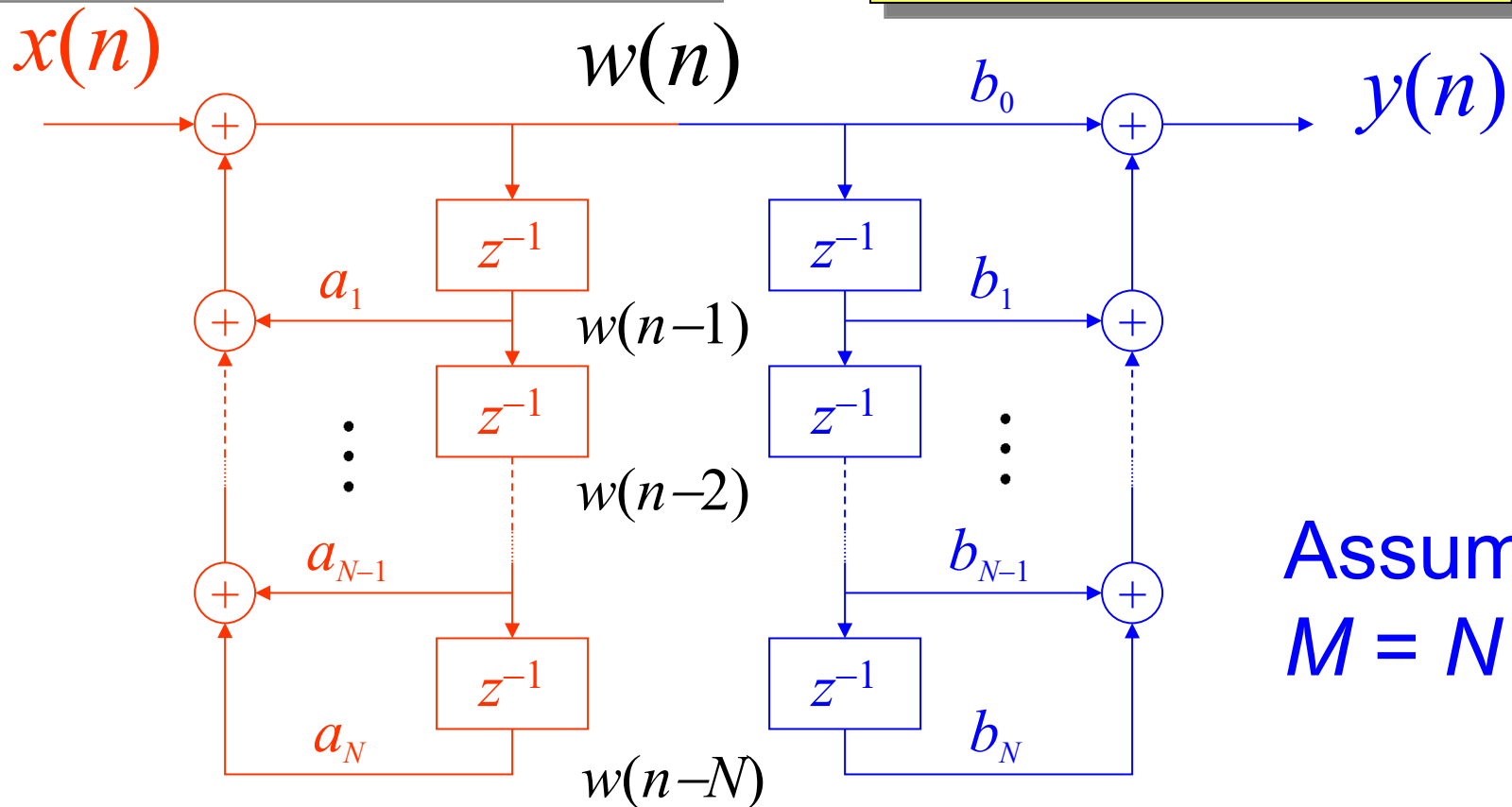


- Block Diagram Representation (Direct Form II) (**Canonic Form**)



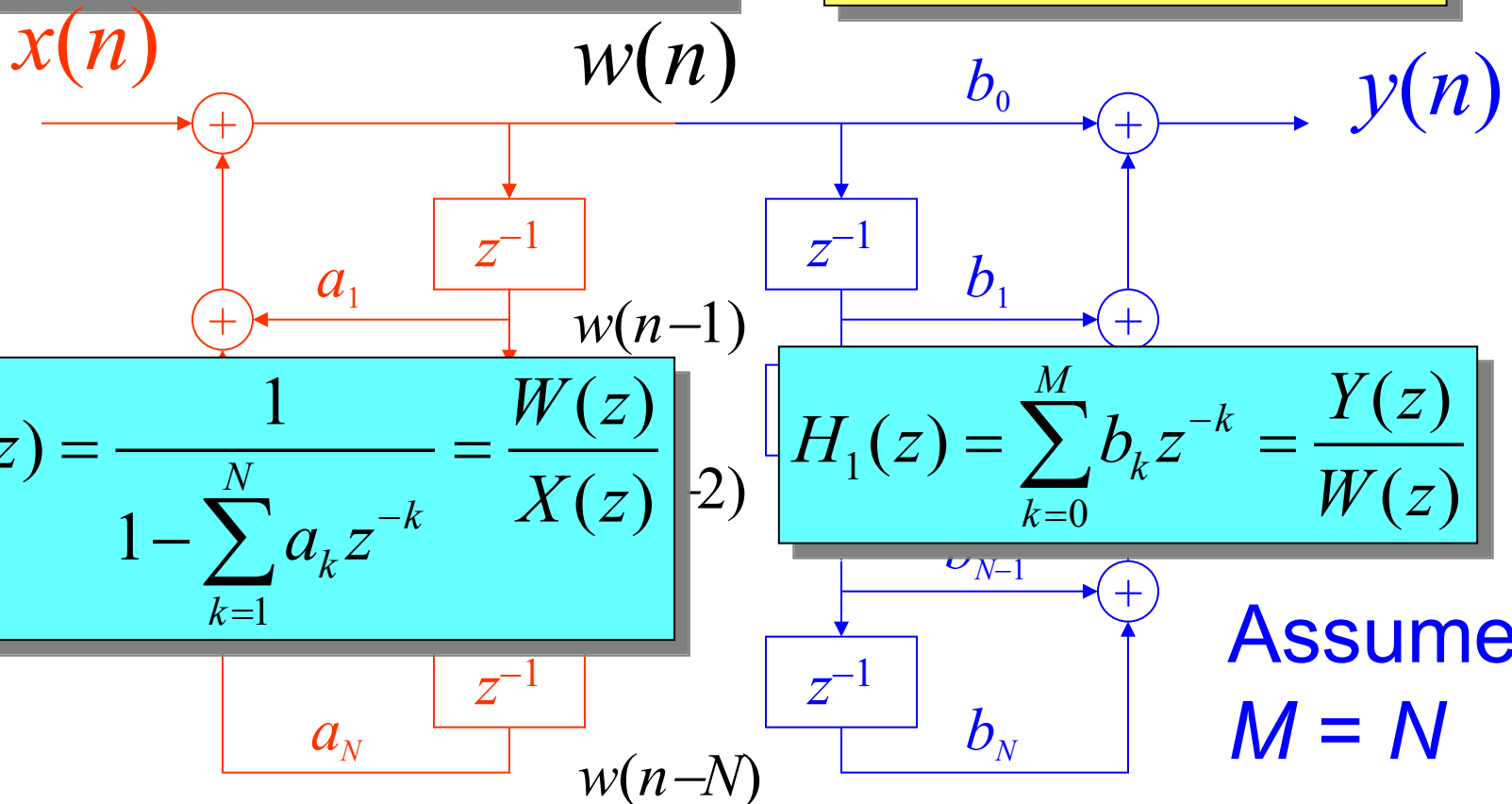
$$w(n) = \sum_{k=1}^N a_k w(n-k) + x(n)$$

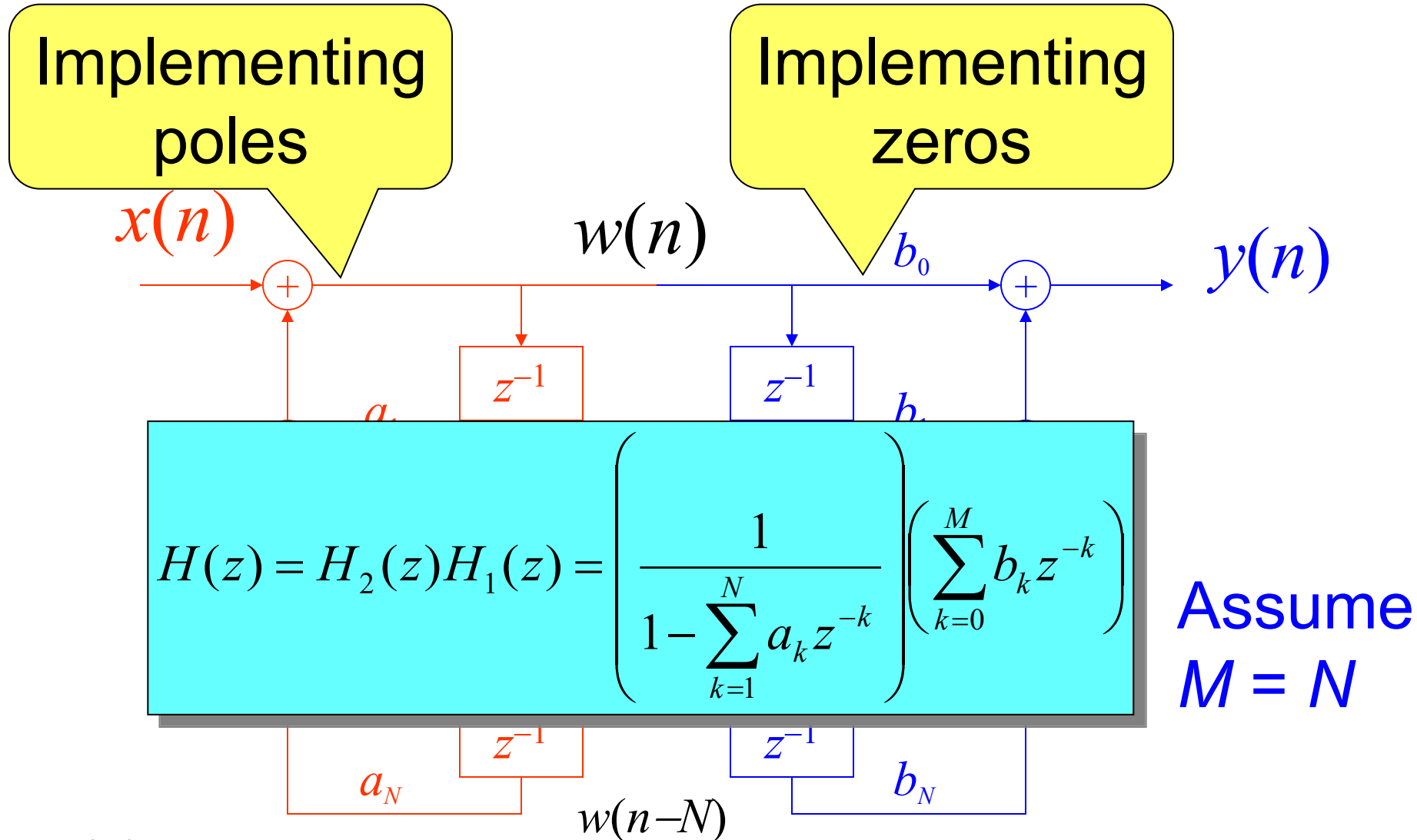
$$y(n) = \sum_{k=0}^M b_k w(n-k)$$



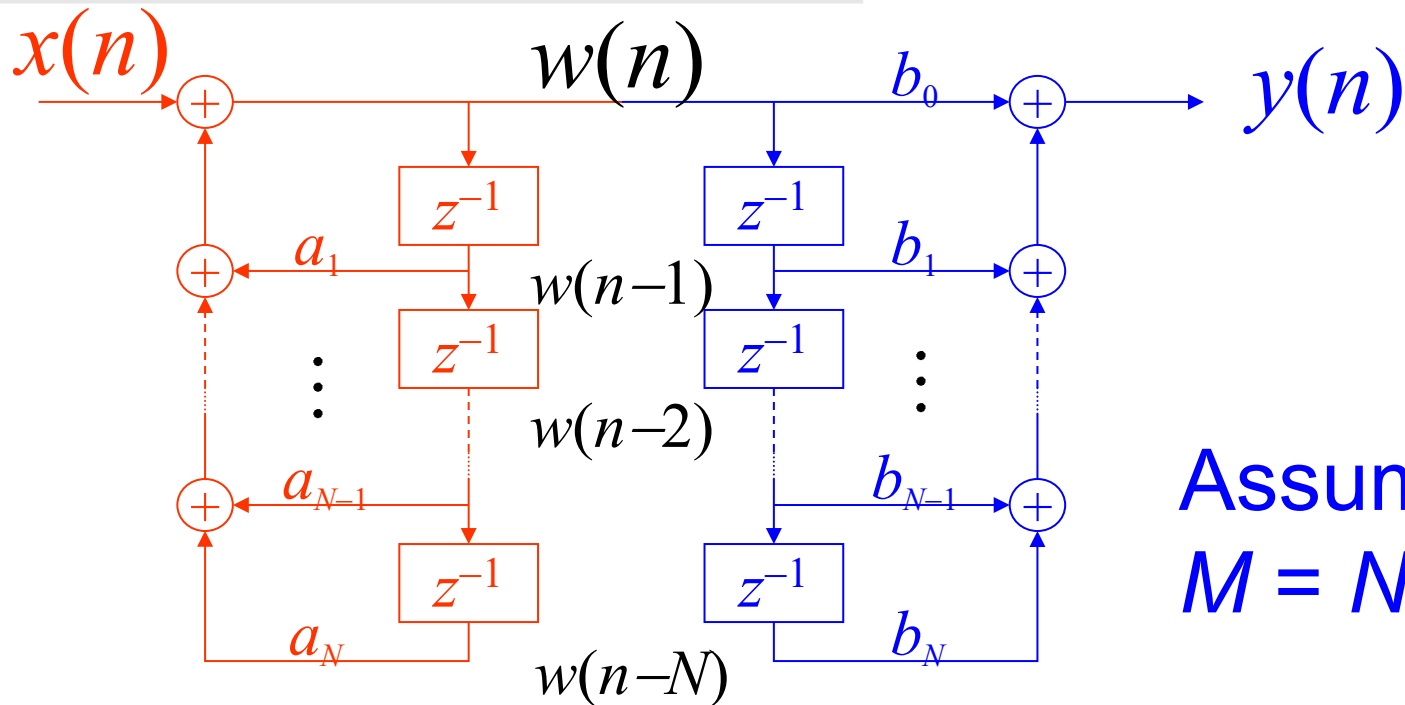
$$w(n) = \sum_{k=1}^N a_k w(n-k) + x(n)$$

$$y(n) = \sum_{k=0}^M b_k w(n-k)$$



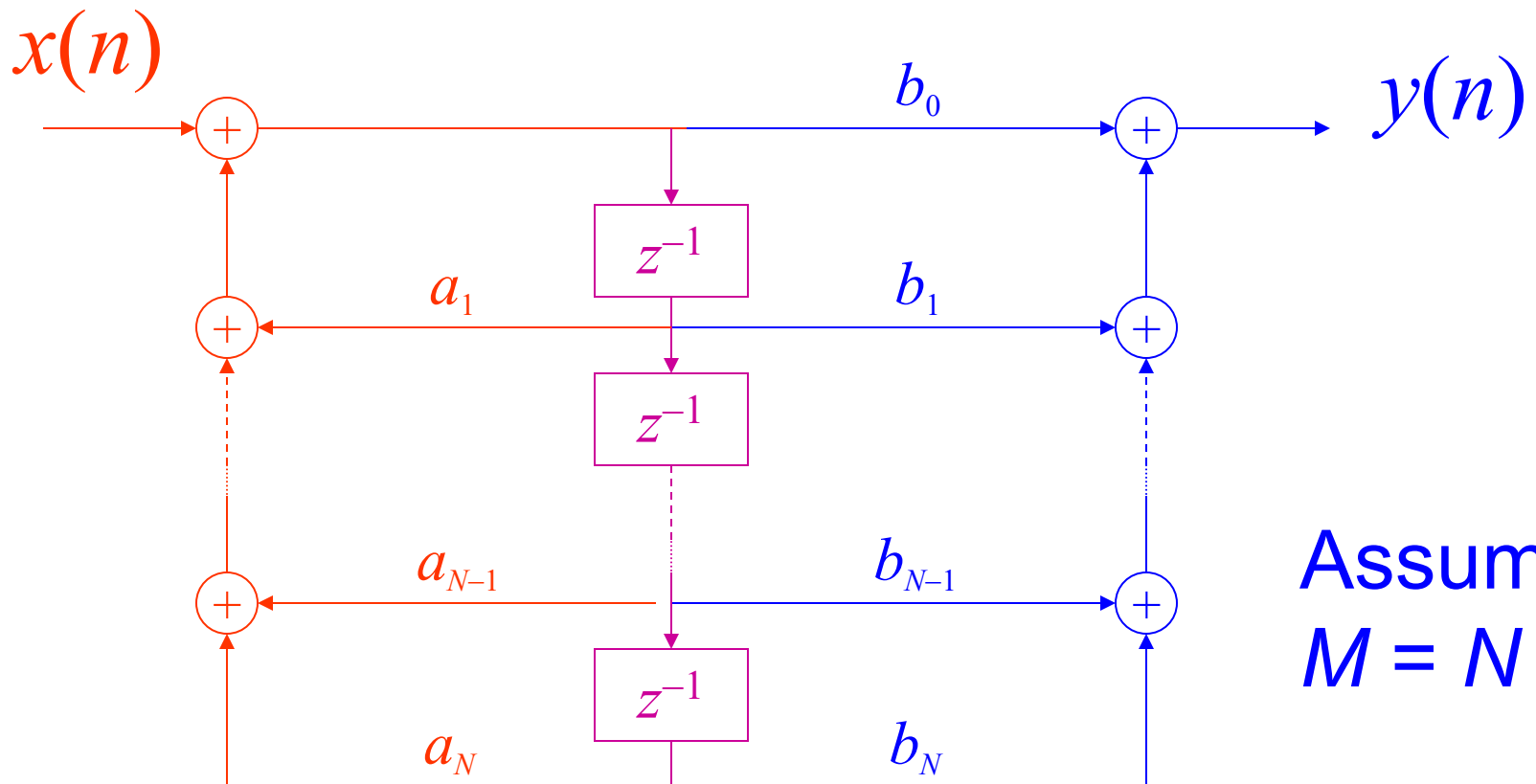


How many Adders? $N + M$
 How many multipliers? $N + M + 1$
 How many delays? $N + M$



Assume
 $M = N$

- Block Diagram Representation (**Canonic Direct Form**)



Assume
 $M = N$

How many Adders?

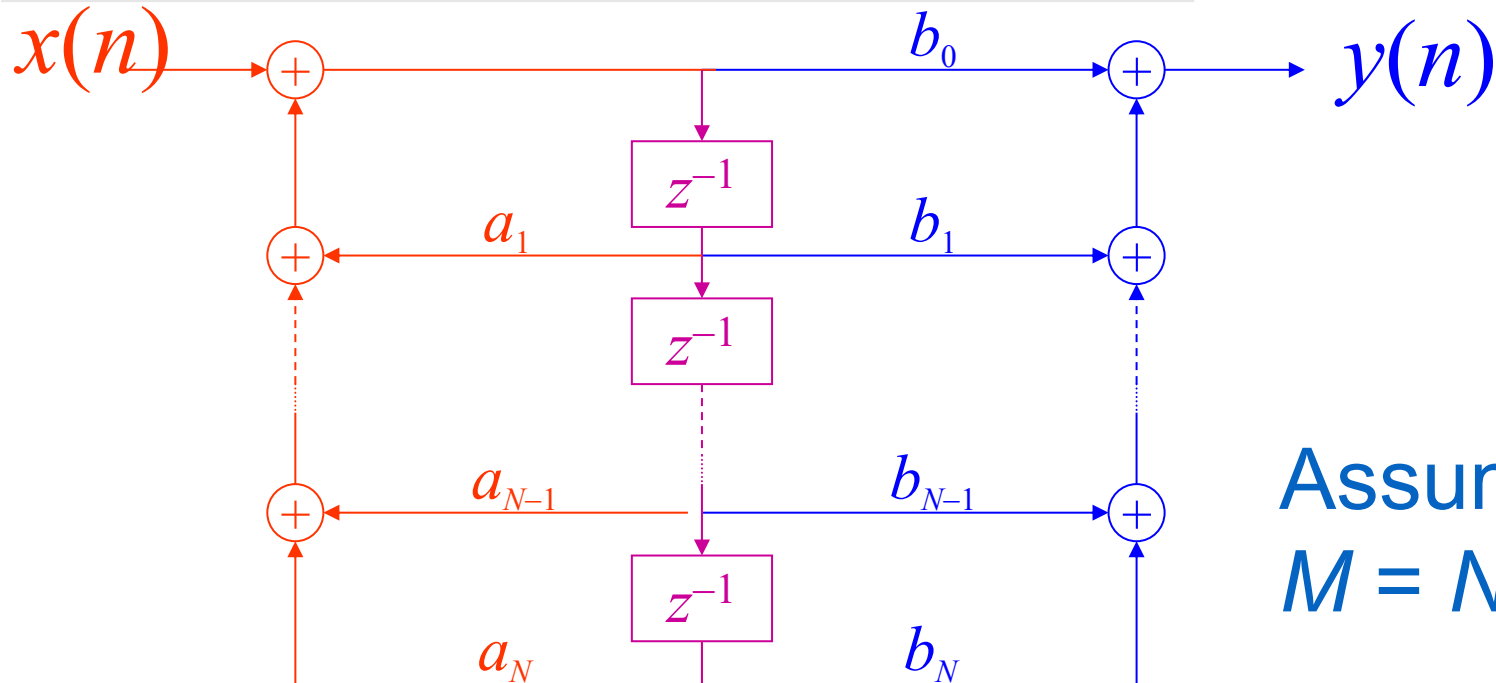
How many multipliers?

How many delays? $\max(M, N)$

$$N + M$$

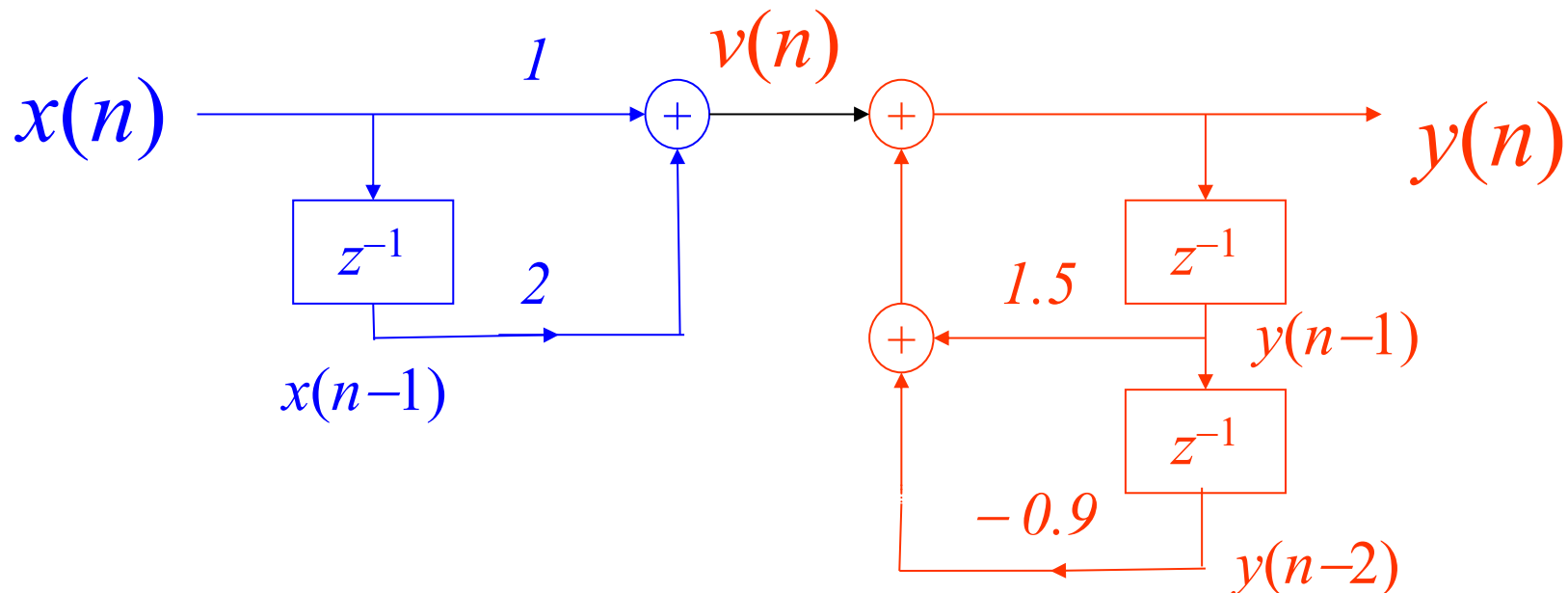
$$N + M + 1$$

$$N$$

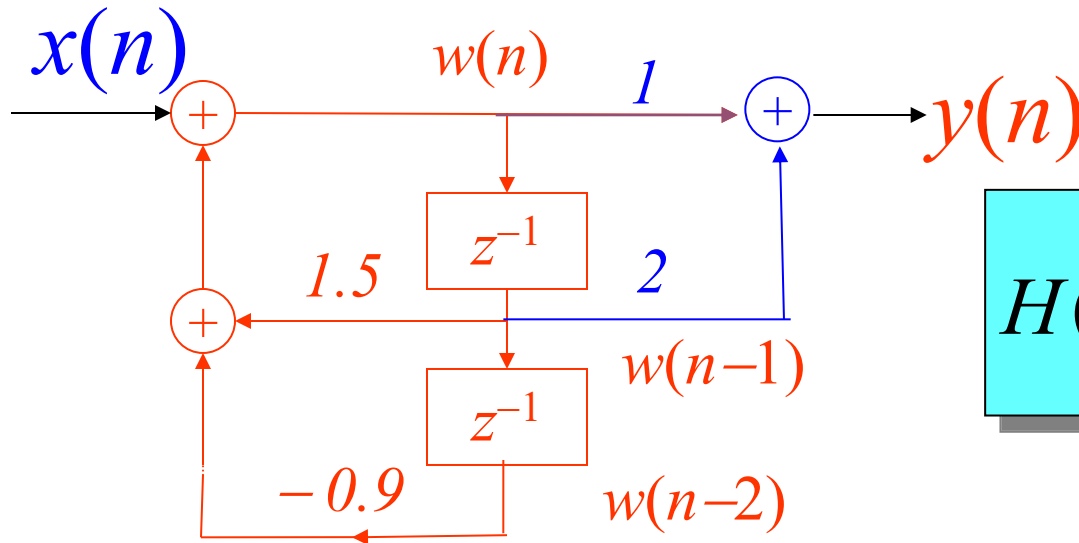


Assume
 $M = N$

- Example: Draw Direct Form I and Direct Form II implementation of an LTI system



• Solution:



$$H(z) = \frac{1 + 2z^{-1}}{1 - 1.5z^{-1} + 0.9z^{-2}}$$

1. Block Diagram Representation of Linear Constant-Coefficient Difference Equations

2. Basic Structures for IIR Systems

3. Basic Structures for FIR Systems

4. The Effects of Coefficient Quantization

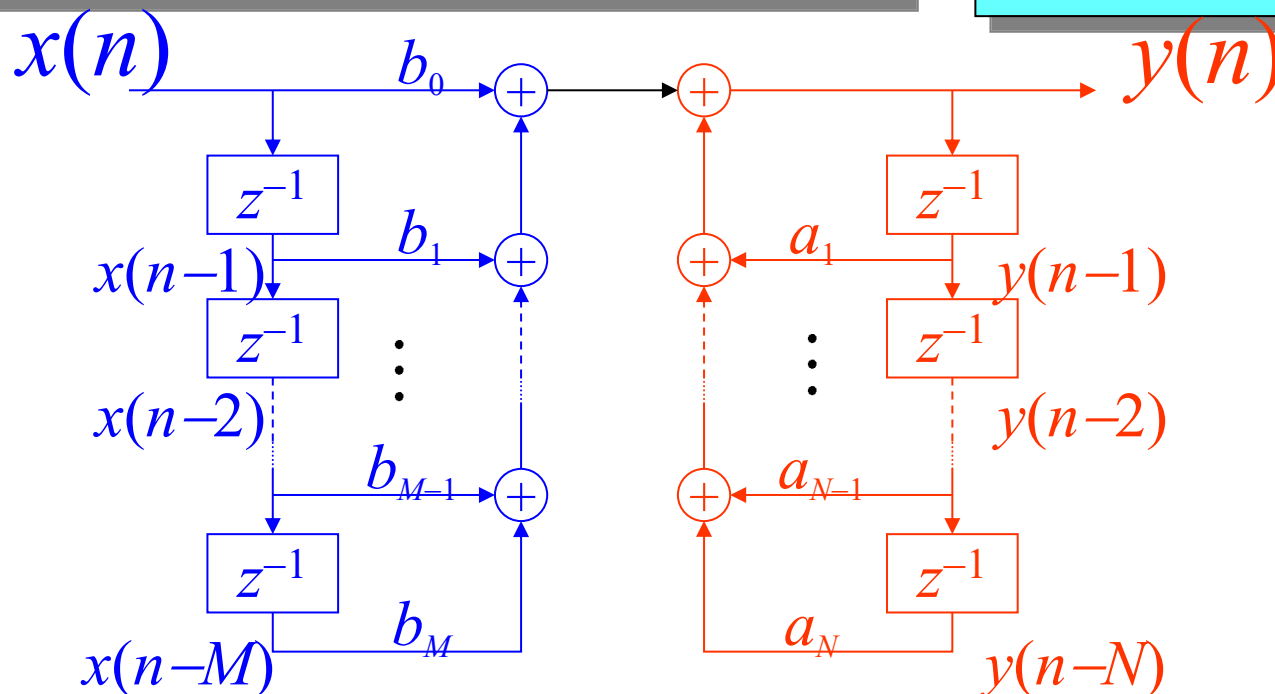
- ◆ For a rational system function, many equivalent difference equations or network structures exists. **Criteria** in the choice among these different structures is computational complexity.
- Reduce the number of constant multipliers
 - Increase speed
- Reduce the number of delays
 - Reduce the memory requirement
- Modularity: VLSI design

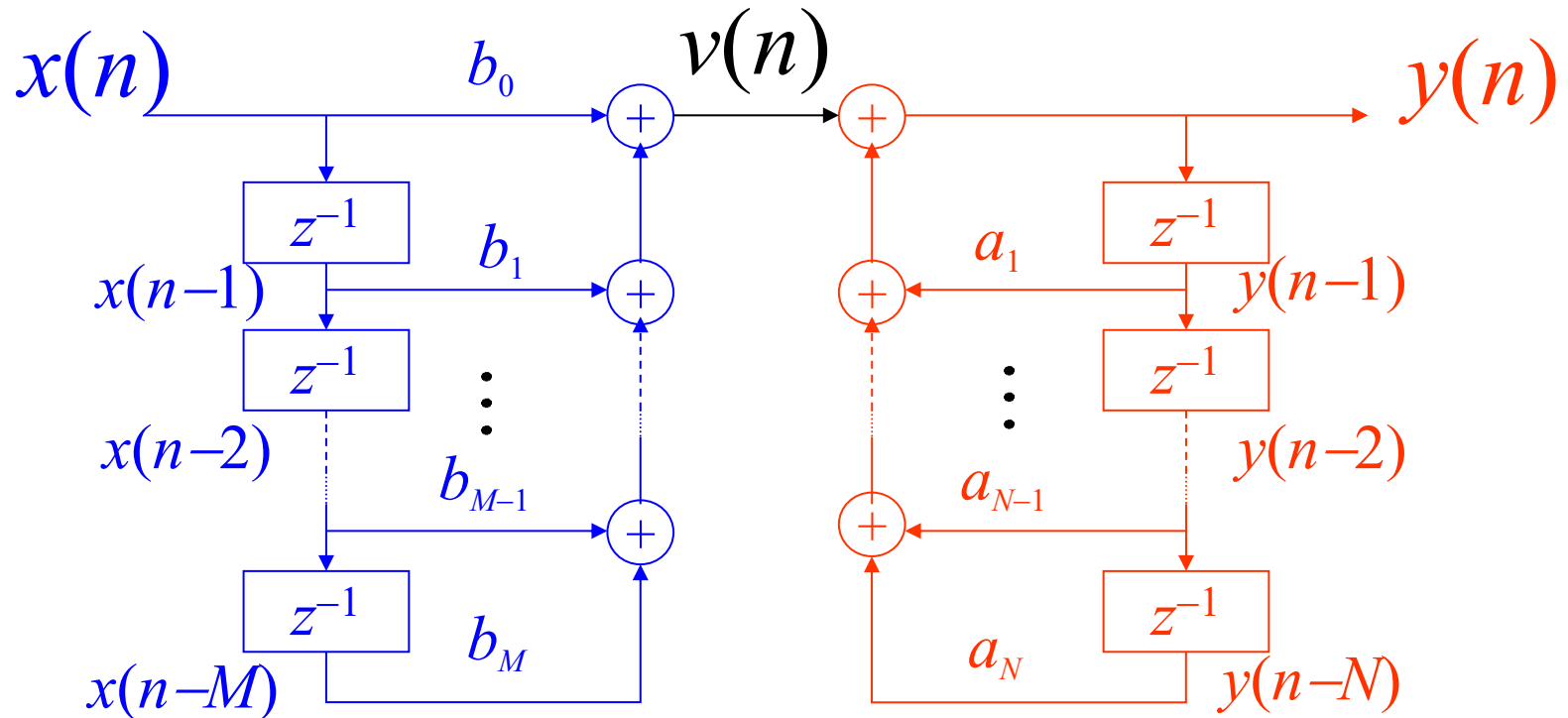
- Basic Structures for IIR Systems
 - Direct Forms
 - Cascade Form
 - Parallel Form

- Direct Forms I

$$y(n) - \sum_{k=1}^N a_k y(n-k) = \sum_{k=0}^M b_k x(n-k)$$

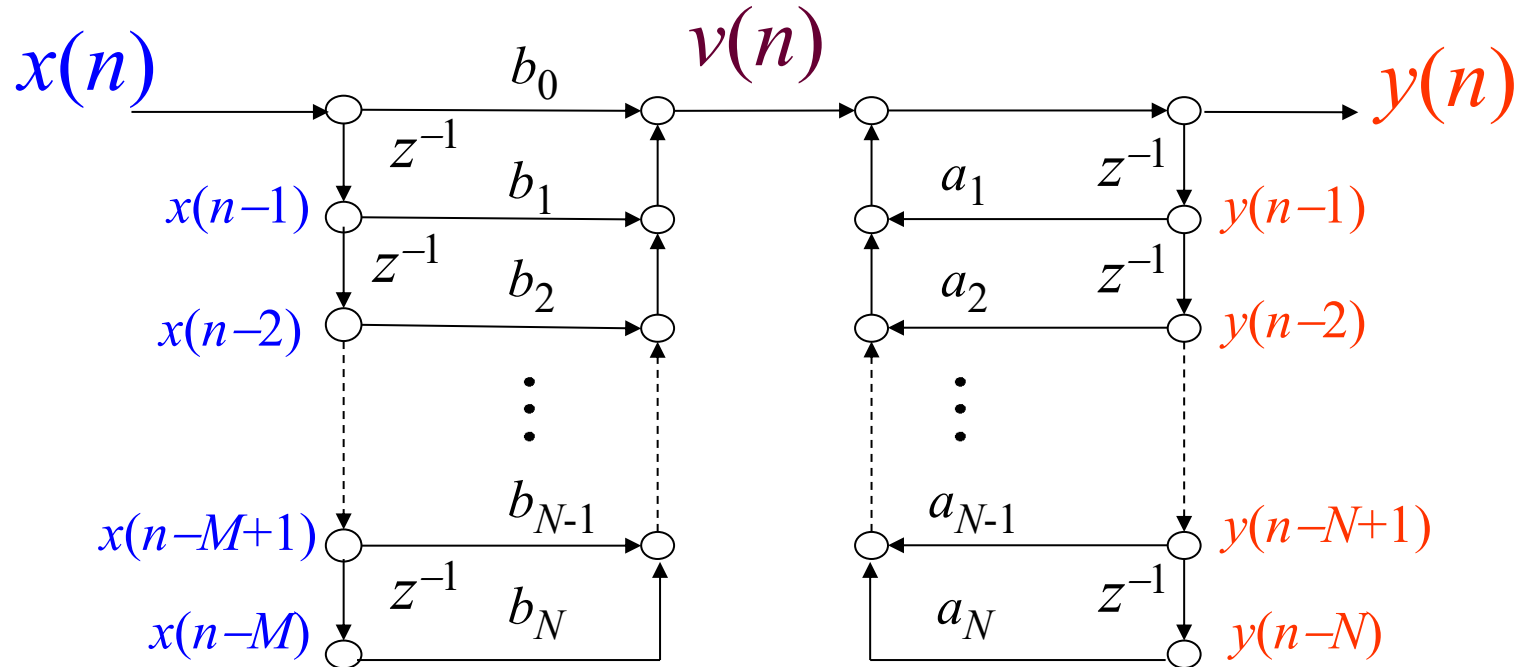
$$H(z) = \frac{\sum_{k=1}^M b_k z^{-k}}{1 - \sum_{k=1}^N a_k z^{-k}}$$





Direct Form I

Block Diagram

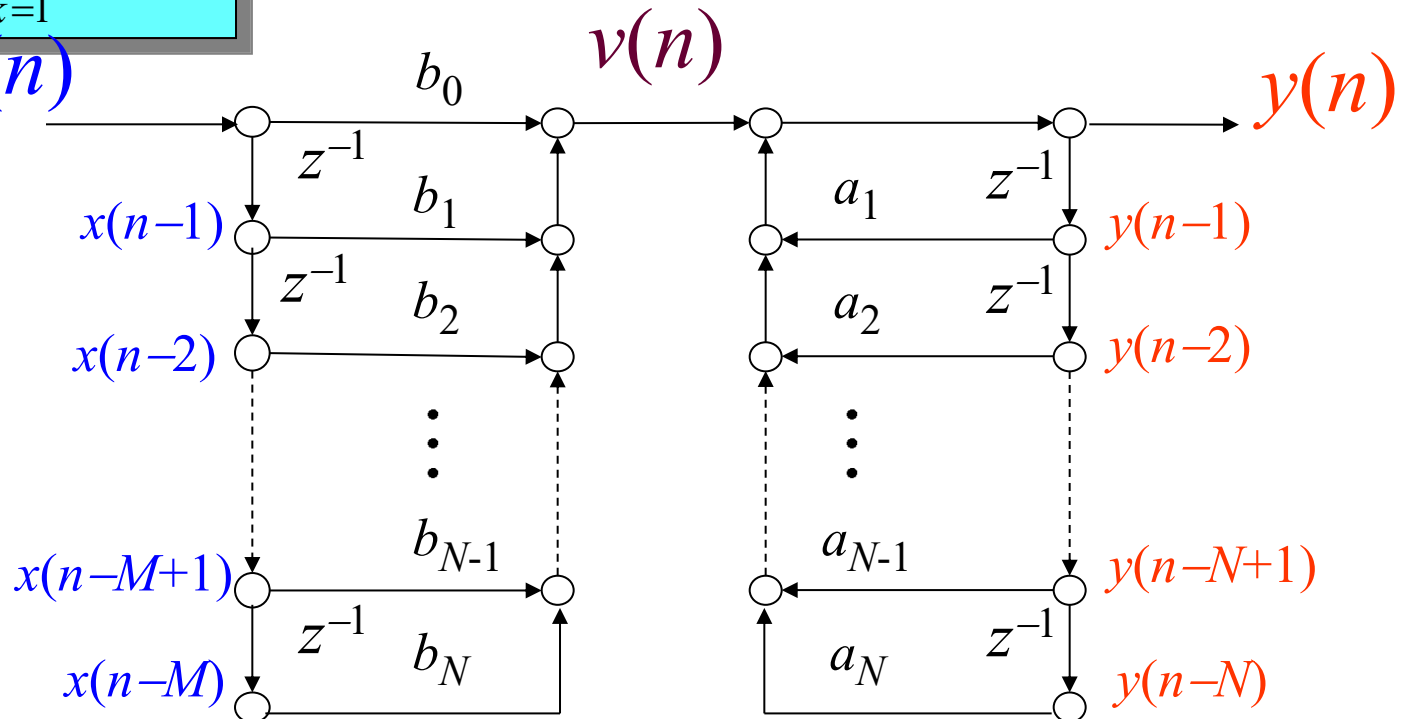


**Signal Flow
Graph**

$$H(z) = \frac{\sum_{k=1}^M b_k z^{-k}}{1 - \sum_{k=1}^N a_k z^{-k}}$$

$x(n)$

$$y(n) - \sum_{k=1}^N a_k y(n-k) = \sum_{k=0}^M b_k x(n-k)$$

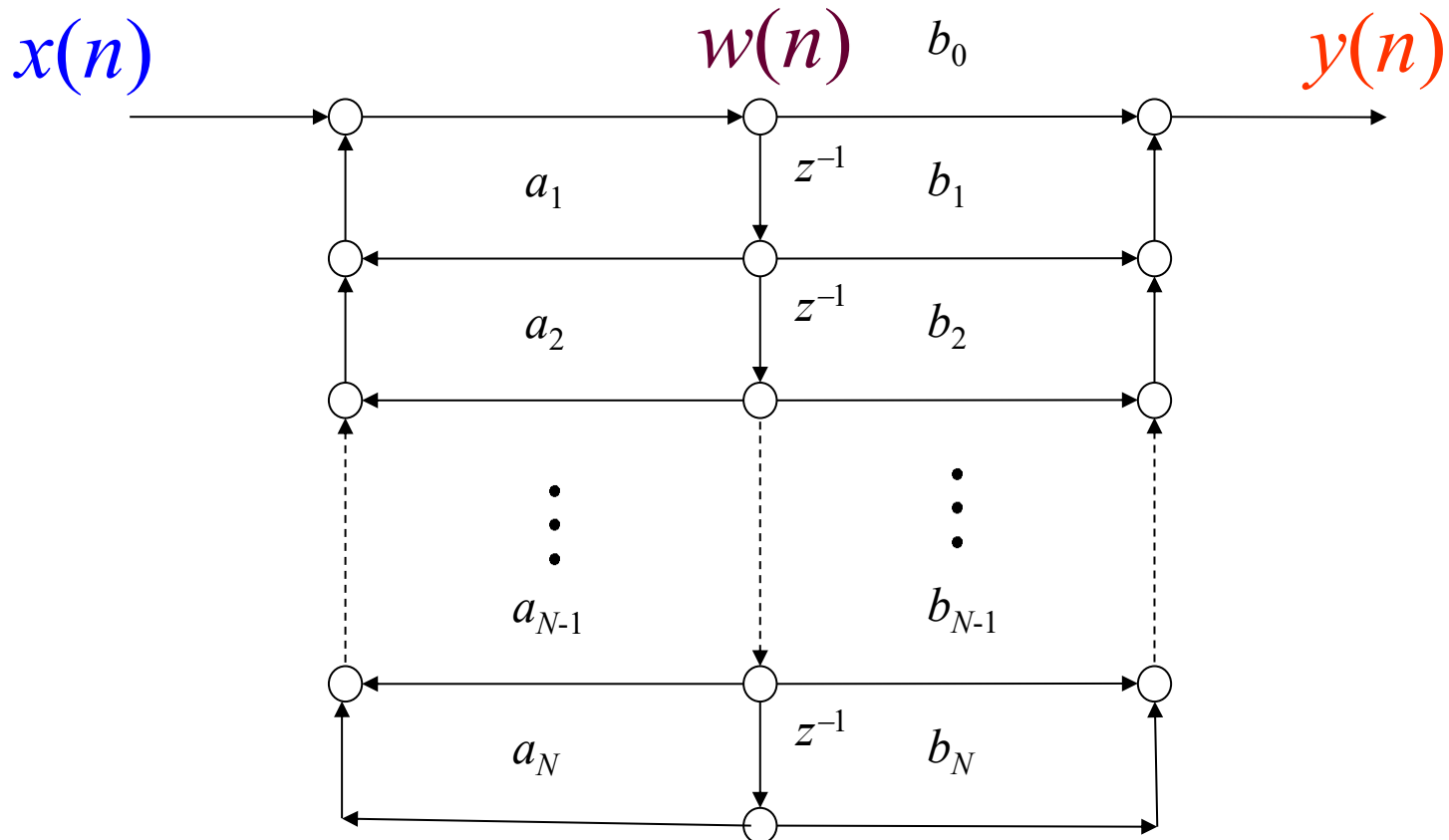


- Direct Form II

$$y(n) - \sum_{k=1}^N a_k y(n-k) = \sum_{k=0}^N b_k x(n-k)$$

$$w(n) = \sum_{k=1}^N a_k w(n-k) + x(n)$$

$$y(n) = \sum_{k=0}^M b_k w(n-k)$$

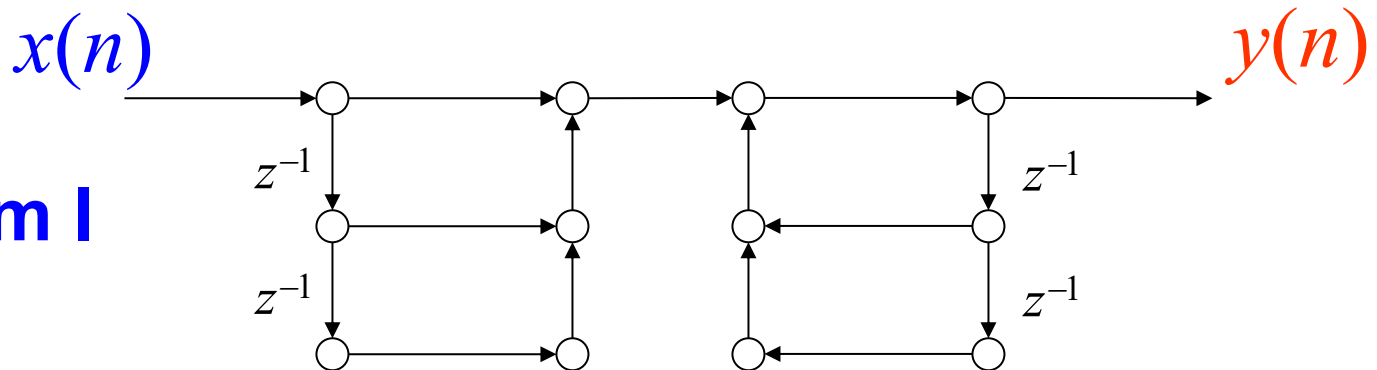


- Example: Draw **Direct Form I** and **Direct Form II** structures of system

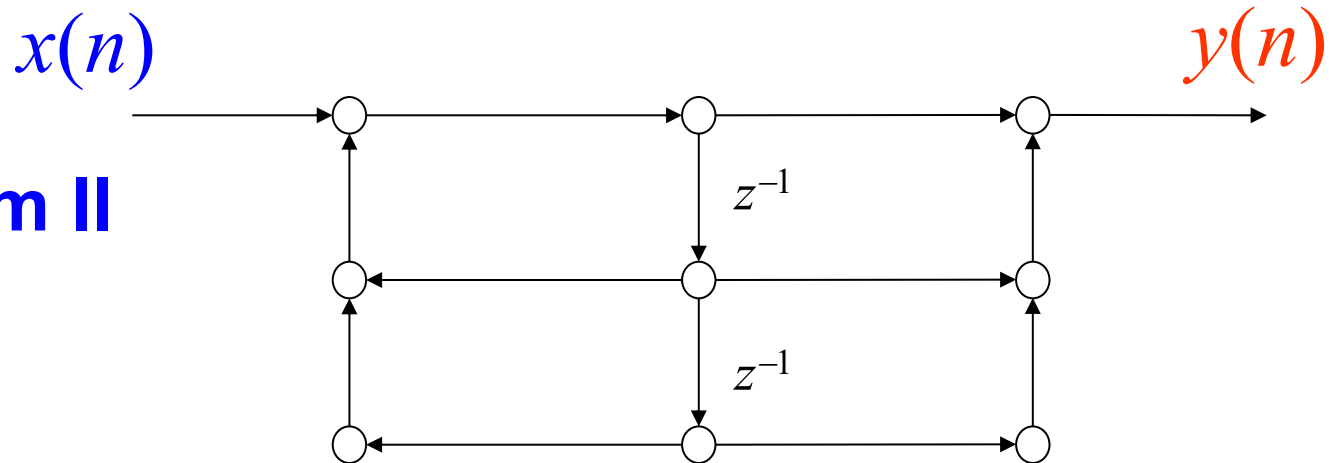
$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

Solution:

Direct Form I



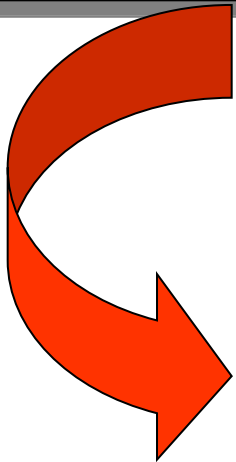
Direct Form II



- Cascade Form

$$y(n) - \sum_{k=1}^N a_k y(n-k) = \sum_{k=0}^M b_k x(n-k)$$

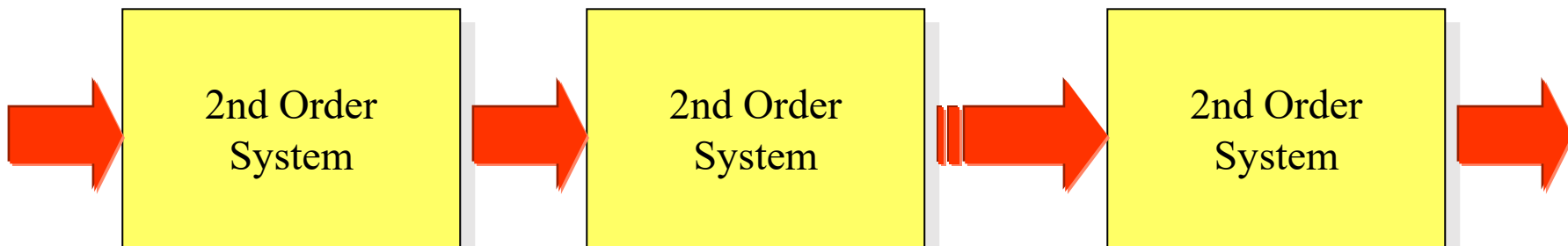
$$H(z) = \frac{\sum_{k=1}^M b_k z^{-k}}{1 - \sum_{k=1}^N a_k z^{-k}}$$



$$H(z) = \frac{\prod_{k=1}^{M_1} (1 - g_k z^{-1}) \prod_{k=1}^{M_2} (1 - h_k z^{-1})(1 - h_k^* z^{-1})}{\prod_{k=1}^{N_1} (1 - c_k z^{-1}) \prod_{k=1}^{N_2} (1 - d_k z^{-1})(1 - d_k^* z^{-1})}$$

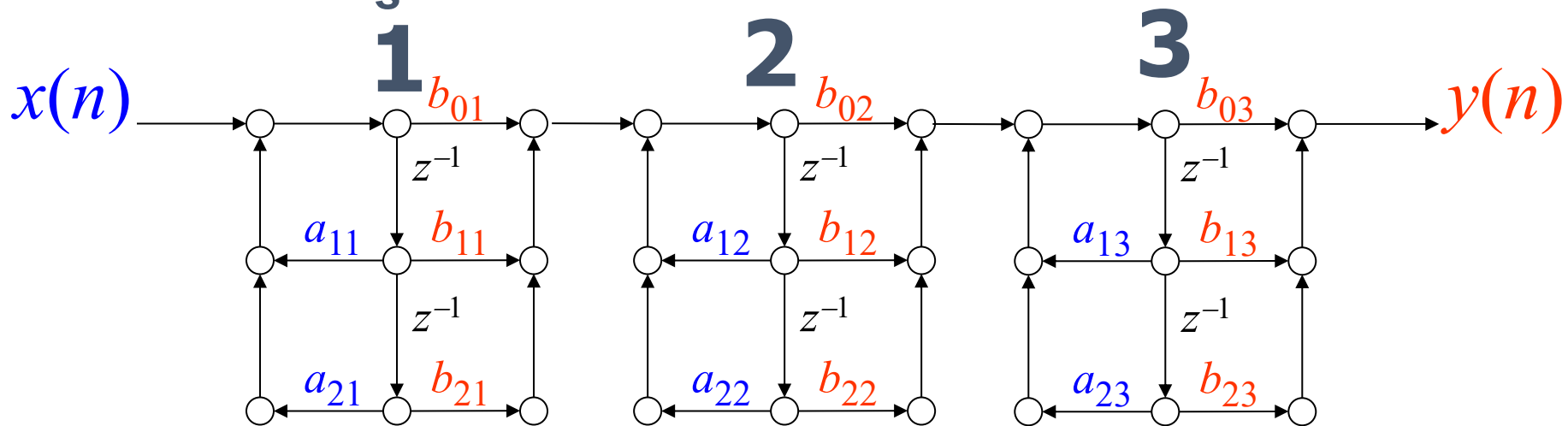
$$H(z) = \frac{\prod_{k=1}^{M_1} (1 - g_k z^{-1}) \prod_{k=1}^{M_2} (1 - h_k z^{-1})(1 - h_k^* z^{-1})}{\prod_{k=1}^{N_1} (1 - c_k z^{-1}) \prod_{k=1}^{N_2} (1 - d_k z^{-1})(1 - d_k^* z^{-1})}$$

$$H(z) = \prod_{k=1}^{N_s} \frac{b_{0k} + b_{1k} z^{-1} + b_{2k} z^{-2}}{1 - a_{1k} z^{-1} - a_{2k} z^{-2}}$$



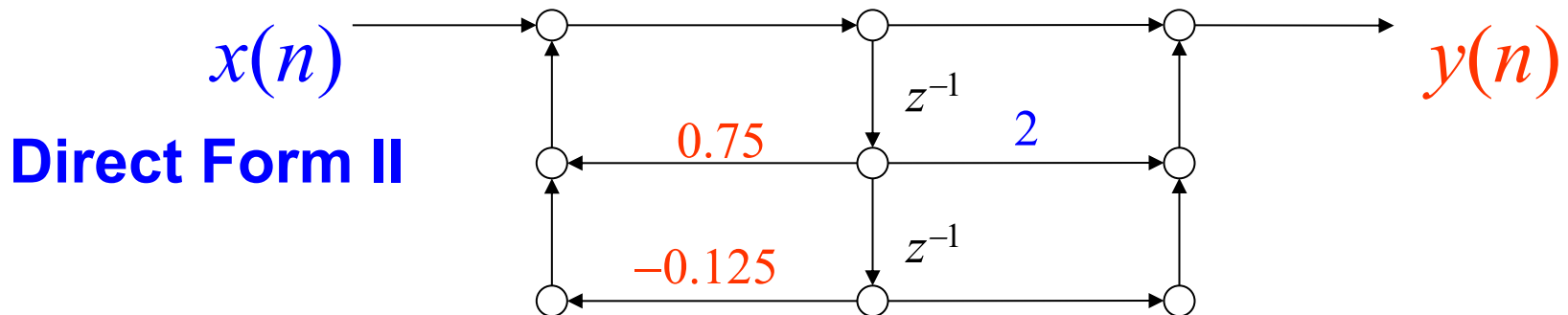
$$H(z) = \prod_{k=1}^{N_s} \frac{b_{0k} + b_{1k}z^{-1} + b_{2k}z^{-2}}{1 - a_{1k}z^{-1} - a_{2k}z^{-2}}$$

Assume $N_s=3$

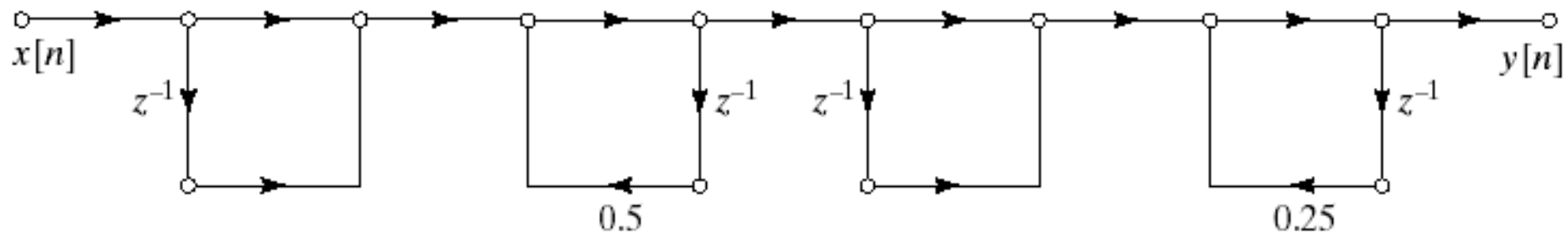


- Example: Illustration of Cascade structures

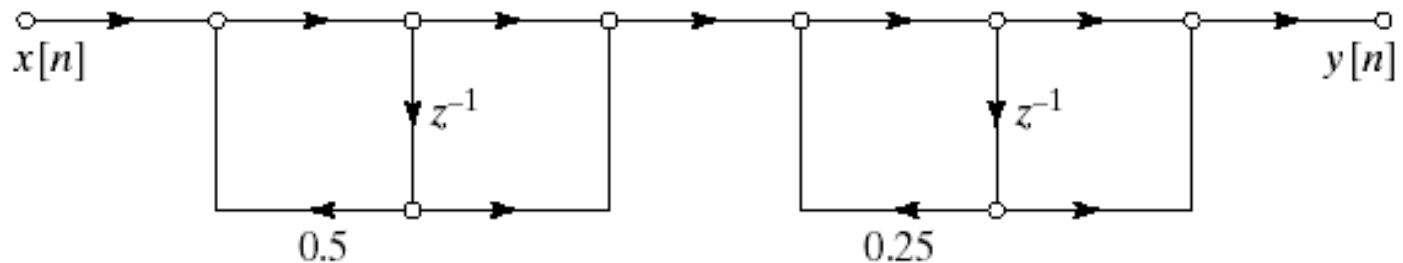
$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}} = \frac{(1 + z^{-1})(1 + z^{-1})}{(1 - 0.5z^{-1})(1 - 0.25z^{-1})}$$



1st-order Direct Form I



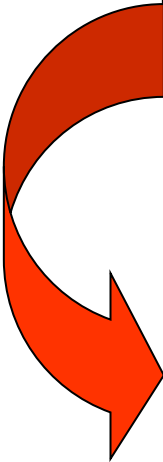
1st-order Direct Form II



- Parallel Form

$$y(n) - \sum_{k=1}^N a_k y(n-k) = \sum_{k=0}^M b_k x(n-k)$$

$$H(z) = \frac{\sum_{k=1}^M b_k z^{-k}}{1 - \sum_{k=1}^N a_k z^{-k}}$$

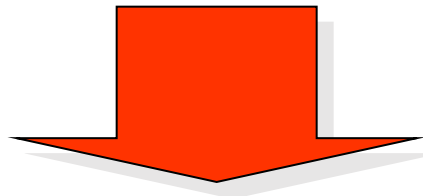


$$H(z) = \sum_{k=0}^{N_P} C_k z^{-k} + \sum_{k=1}^{N_1} \frac{A_k}{1 - c_k z^{-1}} + \sum_{k=1}^{N_1} \frac{B_k (1 - e_k z^{-1})}{(1 - d_k z^{-1})(1 - d_k^* z^{-1})}$$

$$H(z) = \sum_{k=0}^{N_P} C_k z^{-k} + \sum_{k=1}^{N_1} \frac{A_k}{1 - c_k z^{-1}} + \sum_{k=1}^{N_1} \frac{B_k (1 - e_k z^{-1})}{(1 - d_k z^{-1})(1 - d_k^* z^{-1})}$$

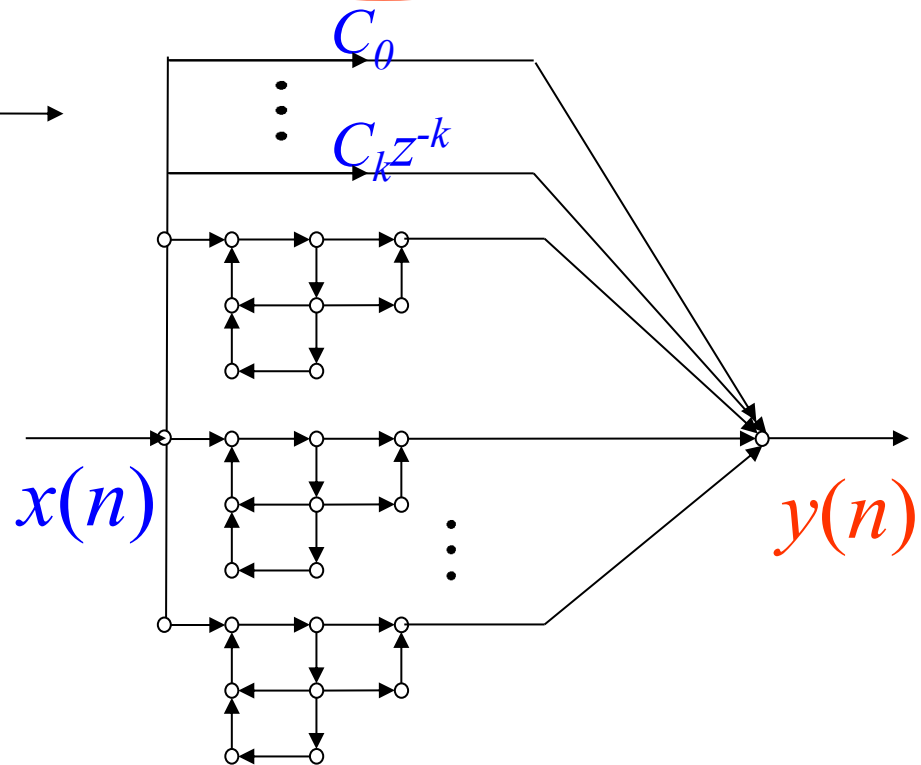
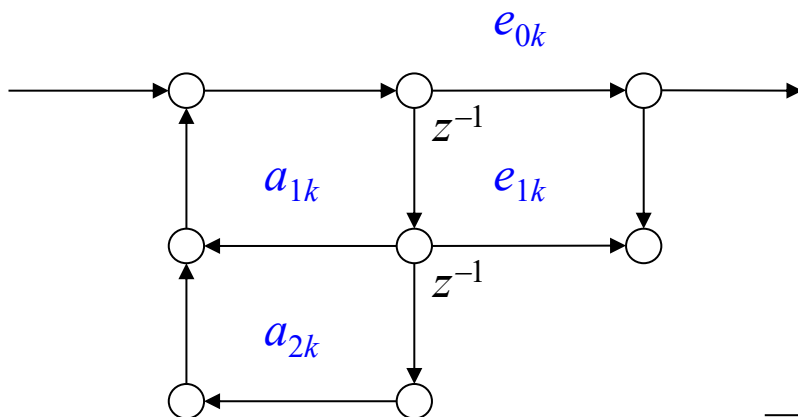
Poles at zero
Real Poles
Complex Poles

Group
Real Poles



$$H(z) = \sum_{k=0}^{N_P} C_k z^{-k} + \sum_{k=1}^{N_s} \frac{e_{0k} - e_{1k} z^{-1}}{1 - a_{1k} z^{-1} - a_{2k} z^{-2}}$$

$$H(z) = \sum_{k=0}^{N_P} C_k z^{-k} + \sum_{k=1}^{N_S} \frac{e_{0k} - e_{1k} z^{-1}}{1 - a_{1k} z^{-1} - a_{2k} z^{-2}}$$

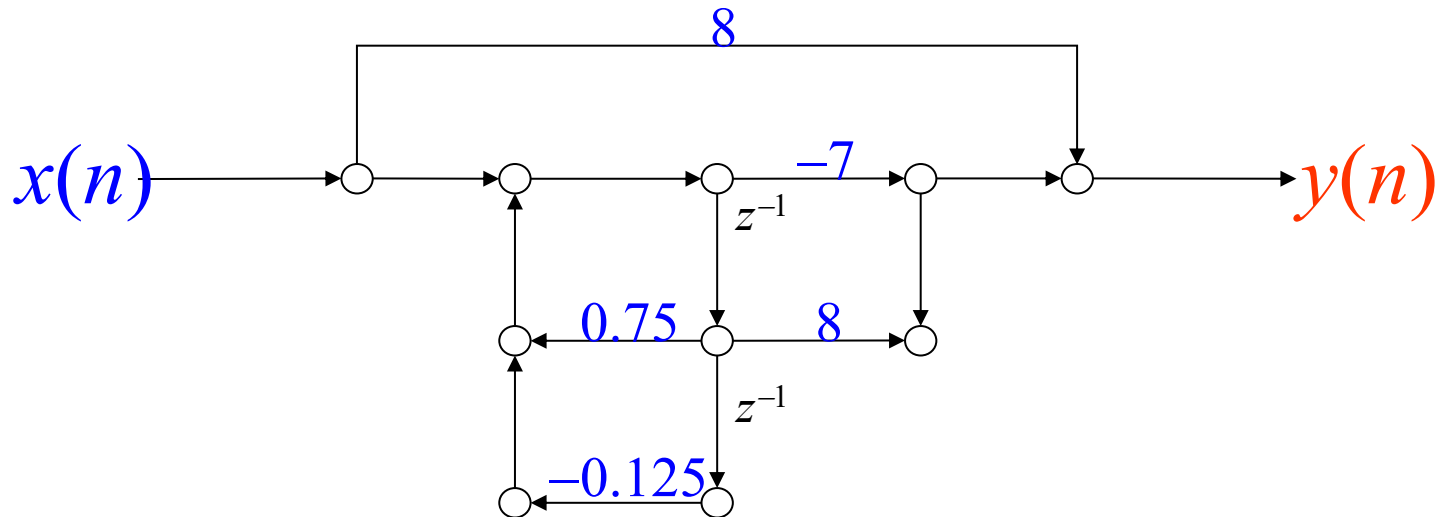


- Example: Draw parallel-form structures of system

$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

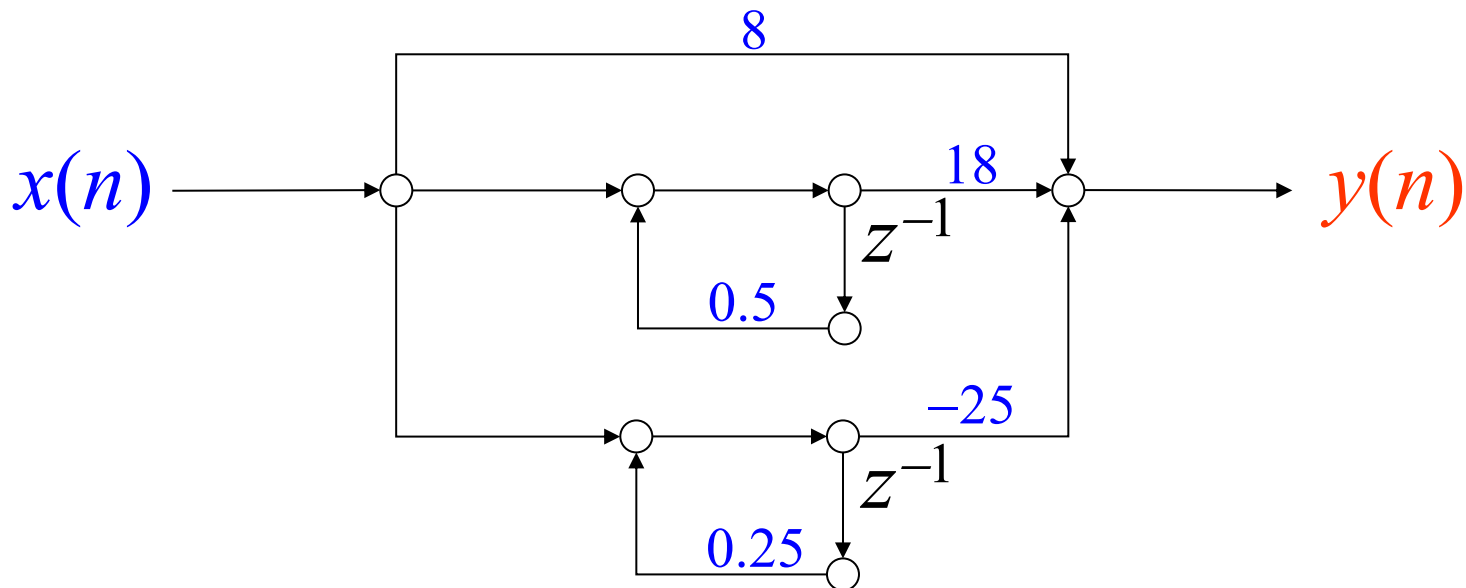
Solution:

$$H(z) = 8 + \frac{-7 + 8z^{-1}}{1 - 0.75z^{-1} + 1.25z^{-2}}$$



$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

$$H(z) = 8 + \frac{18}{1 - 0.5z^{-1}} - \frac{25}{1 - 0.25z^{-1}}$$



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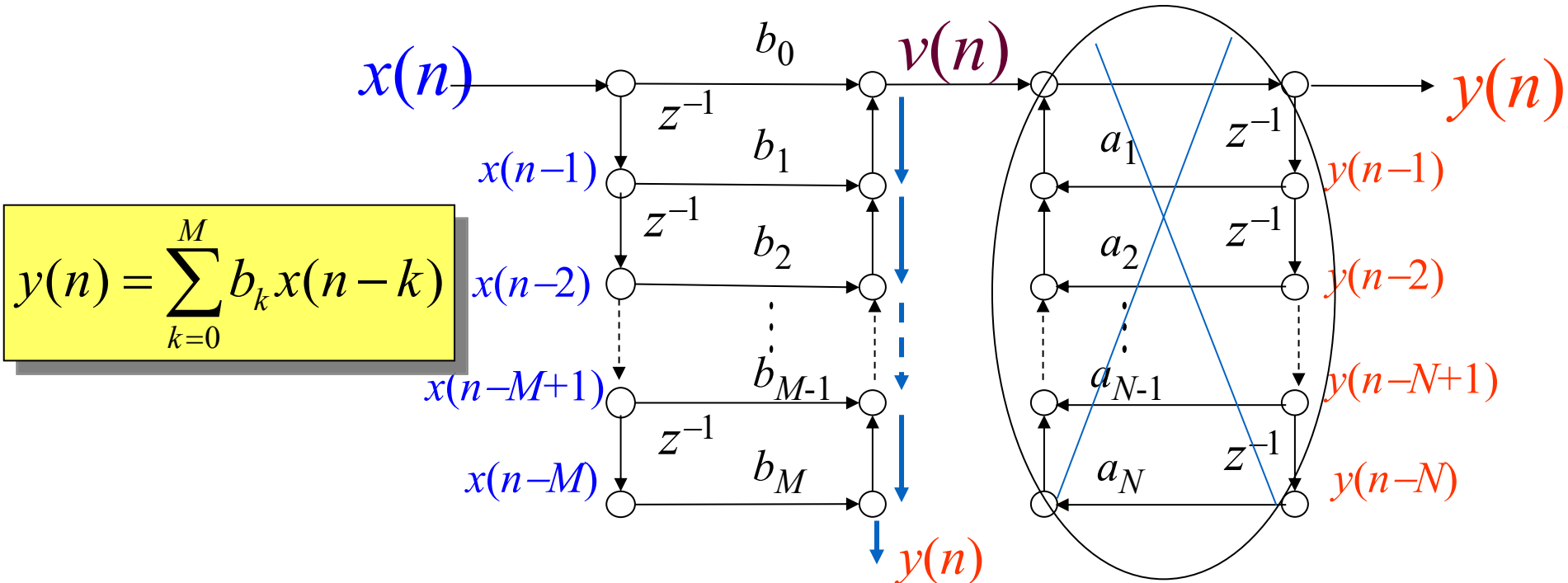
- Direct Form
 - For **causal FIR** systems, the system function has only zeros(except for poles at $z = 0$).

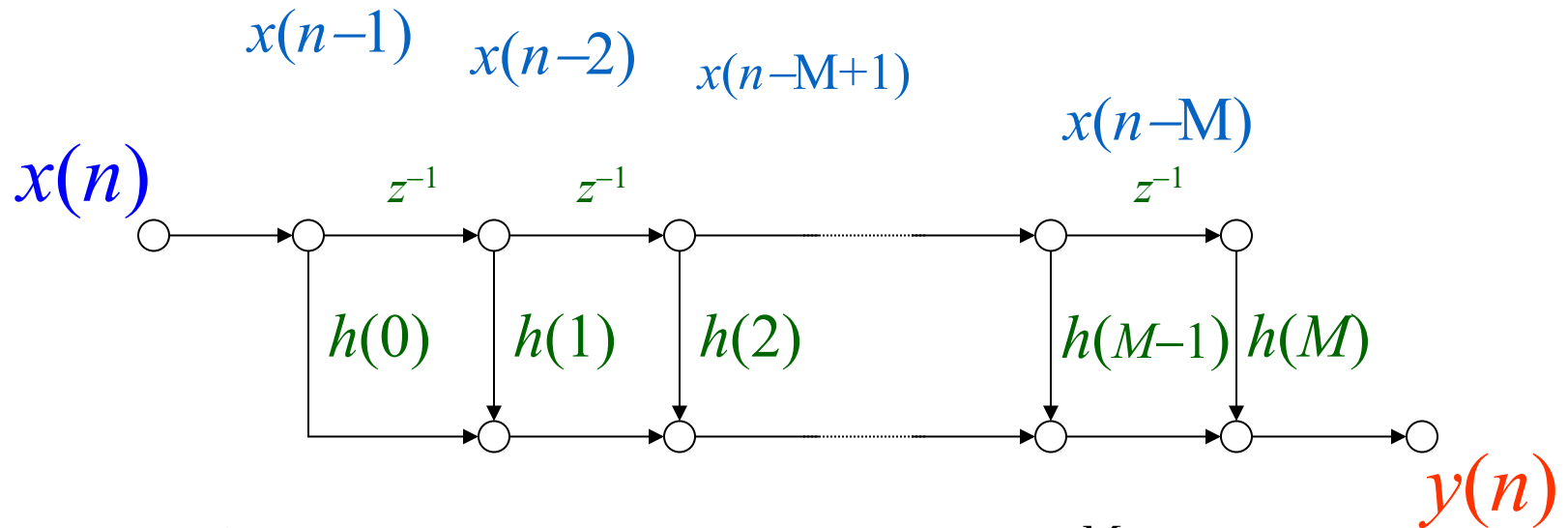
$$y(n) = \sum_{k=0}^M b_k x(n-k)$$

$$y(n) = \sum_{k=0}^M h(k) x(n-k)$$

$$h(n) = \begin{cases} b_n & n = 0, 1, \dots, M \\ 0 & \text{otherwise} \end{cases}$$

- Direct Form I



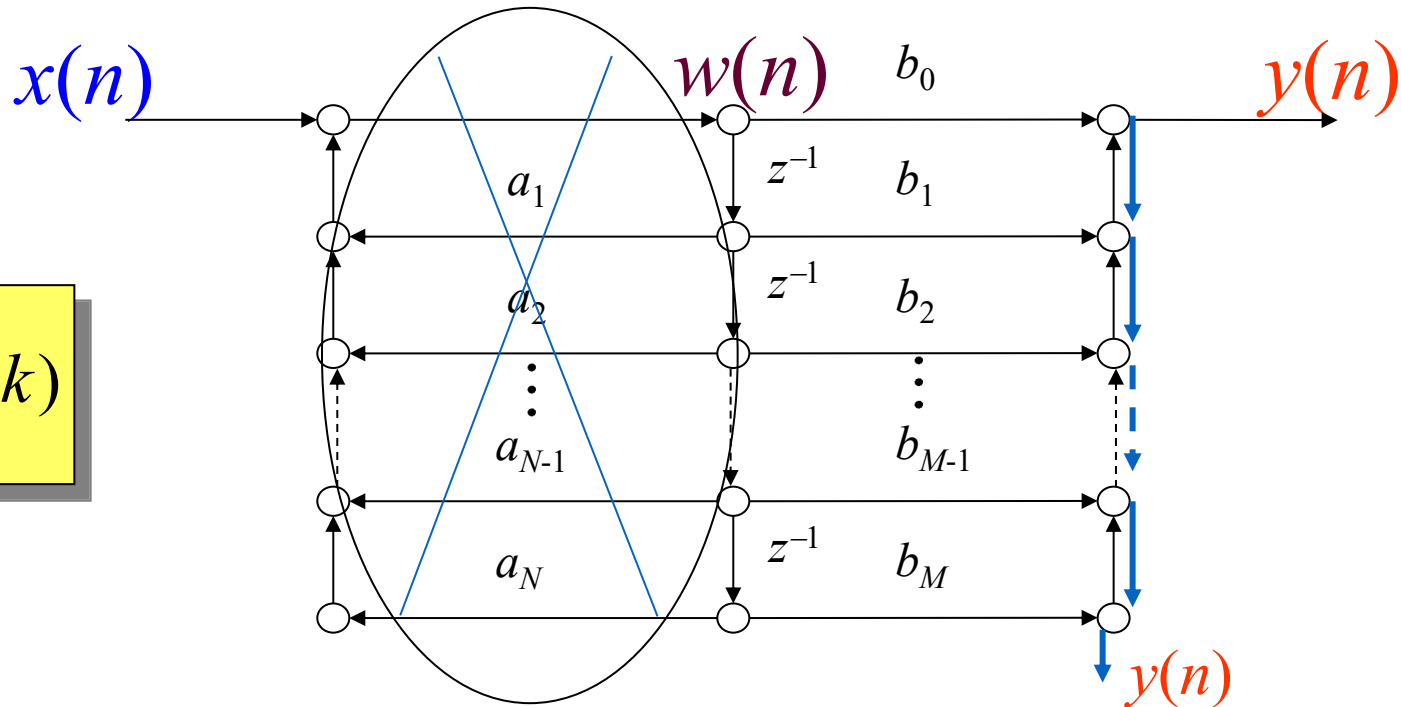


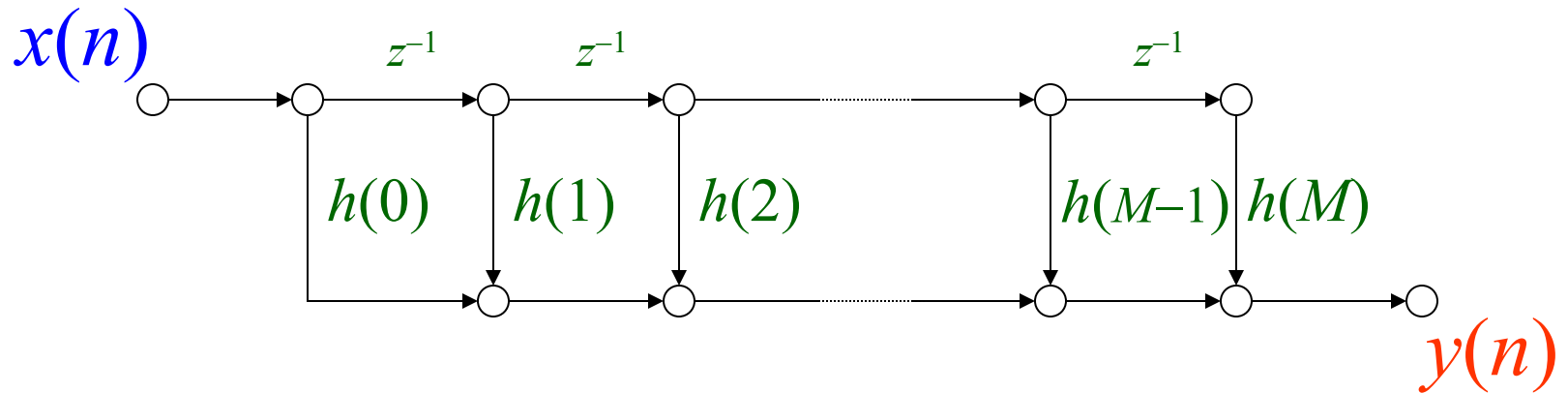
$$h(n) = \begin{cases} b_n & n = 0, 1, \dots, M \\ 0 & \text{otherwise} \end{cases}$$

$$y(n) = \sum_{k=0}^M h(k)x(n-k)$$

- Direct Form II

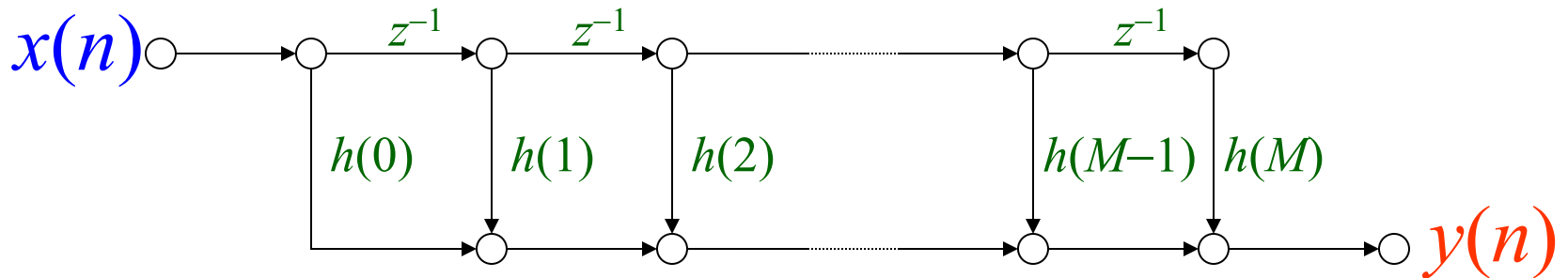
$$y(n) = \sum_{k=0}^M b_k x(n-k)$$



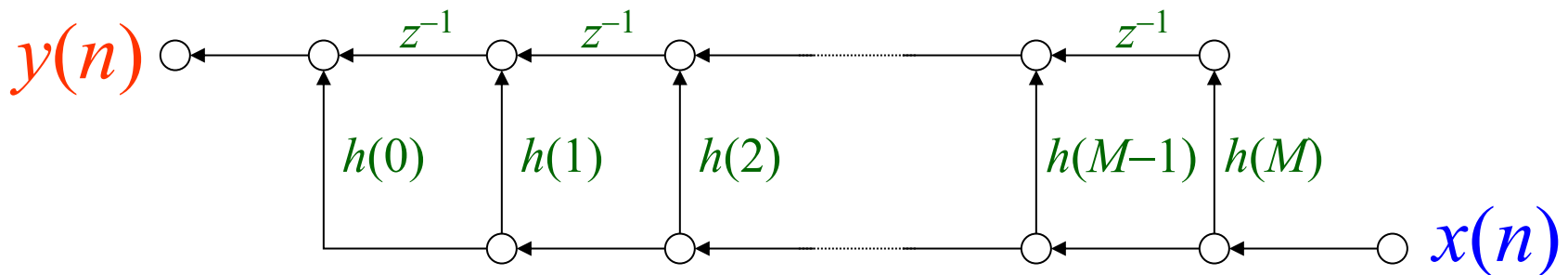


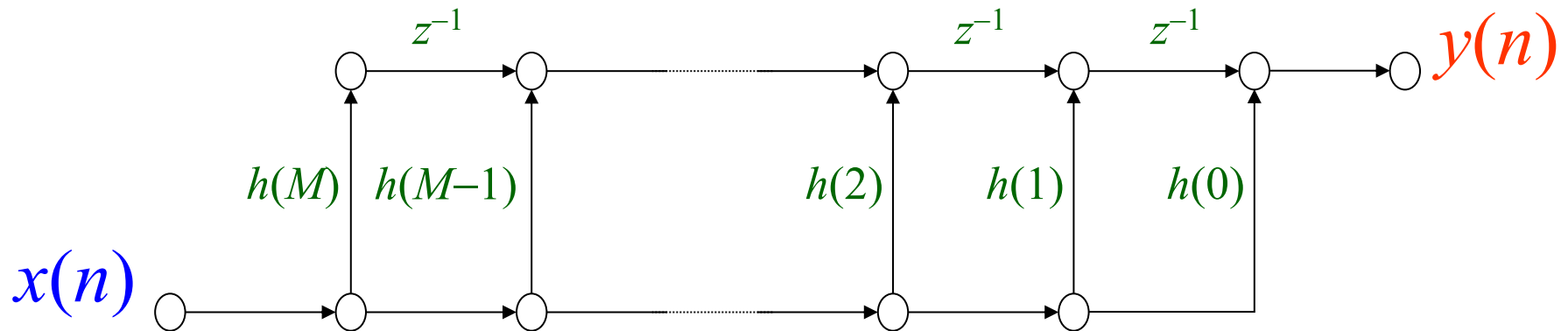
$$h(n) = \begin{cases} b_n & n = 0, 1, \dots, M \\ 0 & \text{otherwise} \end{cases} \quad y(n) = \sum_{k=0}^M h(k)x(n-k)$$

- Transposition of Direct Form



tapped delay line structure or *transversal filter* structure



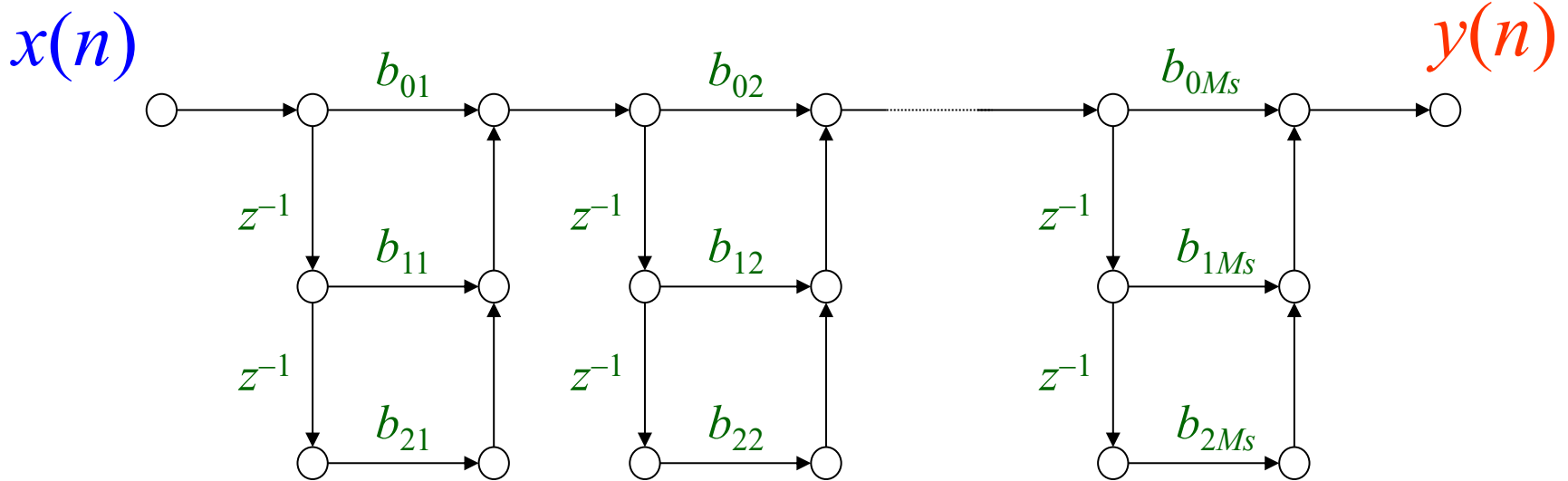


- Cascade Form

$$y(n) = \sum_{k=0}^M b_k x(n-k)$$

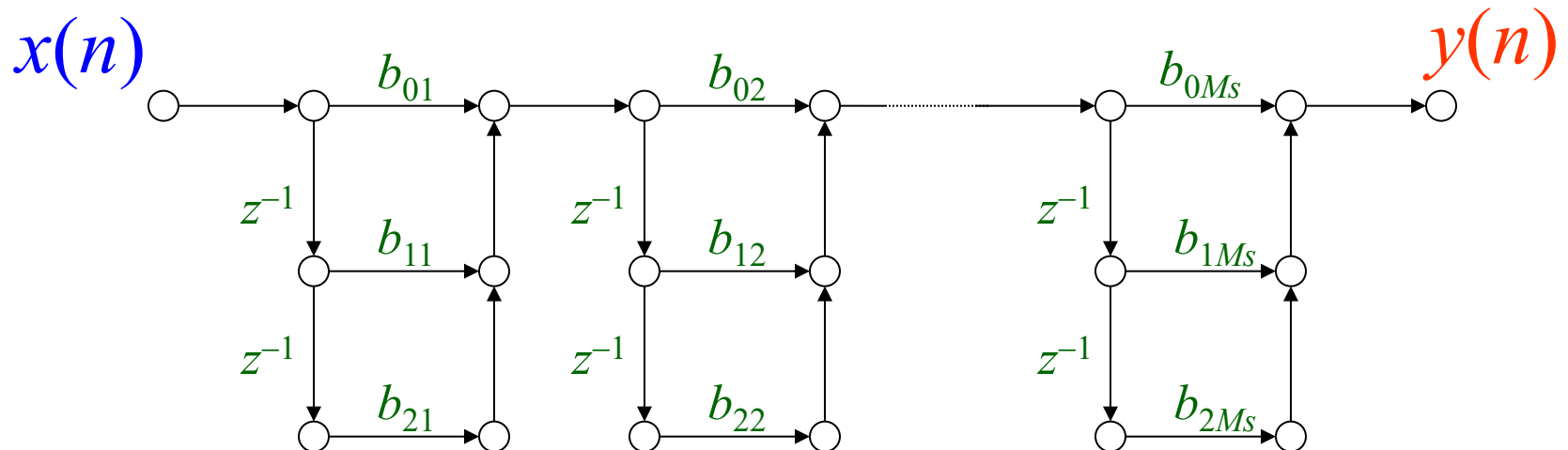
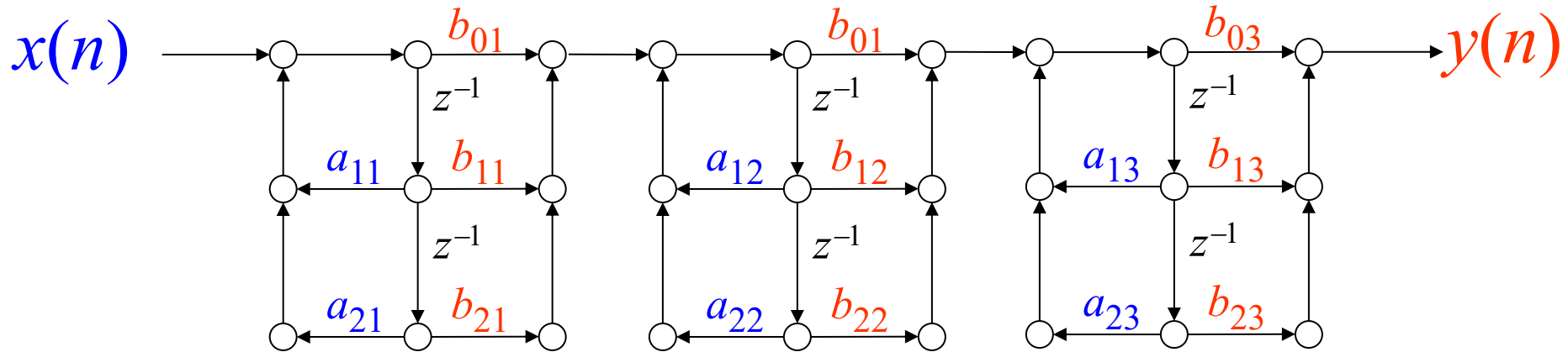
$$H(z) = \sum_{n=0}^M h(n) z^{-n}$$

$$H(z) = \prod_{k=1}^{M_s} (b_{0k} + b_{1k} z^{-1} + b_{2k} z^{-2})$$



$$H(z) = \prod_{k=1}^{N_s} \frac{b_{0k} + b_{1k}z^{-1} + b_{2k}z^{-2}}{1 - a_{1k}z^{-1} - a_{2k}z^{-2}}$$

$$H(z) = \prod_{k=1}^{M_s} (b_{0k} + b_{1k}z^{-1} + b_{2k}z^{-2})$$



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THE EFFECTS OF COEFFICIENT QUANTIZATION

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 - \sum_{k=1}^N a_k z^{-k}}, \quad \hat{H}(z) = \frac{\sum_{k=0}^M \hat{b}_k z^{-k}}{1 - \sum_{k=1}^N \hat{a}_k z^{-k}},$$

$$\hat{a}_k = a_k + \Delta a_k; \hat{b}_k = b_k + \Delta b_k$$

are the quantized coefficients that differ from the original coefficients by the quantization errors

SUMMARY

- This Chapter we have learnt:
 - Many ways to draw block diagram of systems.
 - Design basic structures of IIR and LTI systems.



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THANK YOU